

**Predicting Interdependent Diseases in Individuals
Using Survey Data and Machine Learning Algorithms**

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Abstract

Risk prediction models use explanatory factors in the environment of individuals to generate a probabilistic forecast regarding a health outcome and support decision making in uncertain situations. Facing the COVID-19 pandemic, population level disease transmission modelling is inevitable to plan interventions and evaluate their effectiveness. The risk of the disease underlying the pandemic as well as the occurrence of common diseases can be factors for transmission. Here, an optimized prediction of diseases for the population level is required. In this thesis, optimization possibilities of probabilistic predictions in terms of discriminatory power are investigated. This is done for common diseases on the basis of representative health data for Germany. Special emphasis is placed on the consideration of multiple disease targets and the beneficial use of their interdependence in a multi-label learning framework that predicts all targets simultaneously or sequentially. Based on evaluation metrics, this thesis compares multi-label approaches along with various probabilistic algorithms and finds that the sole use of target dependencies yields only insignificant improvements in predictions. A significant improvement results for ensembles that additionally consider the heterogeneity of algorithms and approaches. The mandatory survey data preprocessing has a general advantageous effect and holds potential for further improvements in predictive performance.

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List of Abbreviations

BC	Binary Cross-Entropy
BR	Binary Relevance
BRS	Binary Relevance Stacking
BS	Brier Score
CART	Classification And Regression Trees
CC	Classifier Chain
CS	Chi-squared Score
CT	Classification Trees
DNN	Deep Neural Network
EBR	Ensemble of Binary Relevance
ECC	Ensemble of Classifier Chains
FCD	Fully Conditional Dependency
FPR	False Positive Rate
GB	Gradient Boosting
GE	Generalization Error
GEDA	German Health Update
HBE	Heterogeneous Bagging Ensemble
HSE	Heterogeneous Stacking Ensemble
LP	Label Powerset
LR	Logistic Regression
MBC	Multi-label Binary Cross-Entropy
MICE	Multiple Imputation by Chained Equations
MLP	Multi-Layer Perceptron
MLRF	Multi-label Random Forest
OHE	One-Hot Encoding
PPV	Positive Predictive Value
PRC	Precision-Recall Curve
RF	Random Forest
ROC	Receiver Operating Characteristic
TPR	True Positive Rate
WoE	Weight-of-Evidence

1 Introduction

The prediction of adverse health outcomes in individuals has a wide range of applications in clinical practice and public health management (Moons et al. (2012), p. 683). It supports decision-making and improves diagnostics, leading to better cost-benefit efficiency in health-care. In the case of disease diagnosis, predictions are used to determine the disease risk. This is expressed as the probability that a individual has the disease at the time of assessment. The probability is, as a measure of risk, determined by explanatory factors in and around the individuals of interest. This is especially relevant for individuals and diseases that are asymptotic. In situations without diagnostic information, prediction allows for proactive and preventive actions. Disease outbreaks and potential adverse outcomes can be addressed in advance and diseases become increasingly manageable.

This enhancement to conventional diagnostics become inevitable in the light of the pandemic outbreak of COVID-19 and the ever since ongoing evolution of new variants of concern as well as the threat of future pandemics in a globalized world. With Germany as population of interest, even a single population-wide diagnosis is infeasible. The responsible authorities can only react to the latest developments without taking anticipatory measures. This puts them one step behind the current pandemic development. The decision-making process can be supported with simulations that help to make pandemics manageable. This is shown by recent agent-based models of Kerr et al. (2021) or Hinch et al. (2021). They assess the optimized use of transmission control measures for COVID-19, ranging from physical distancing to testing and masking to priority vaccination as well as preventive medicine and care. This makes the planning and the evaluation of interventions possible.

The predicted probabilities for common diseases as well as the pandemic disease itself can be used as factors for transmission dynamics. The quality of predicted population probabilities for the relevant diseases becomes one of many complex challenges around the modeling of pandemic dynamics. In order to assess this, a comparative study of disease prediction algorithms on the basis of representative survey data is conducted. Due to the limited availability of COVID-19 data for Germany, only common diseases are considered in this thesis. The data stems from the German Health Update 2014/2015, a health survey for the German population. Here, disease outcomes are available alongside explanatory factors. The representativeness of the data allows for the application of the created algorithms on the German population, where explanatory factors are more readily available than diagnoses. This motivates the creation of an optimized prediction algorithm for information transfer

from the survey to the population level. Based on the survey data, this thesis assess potential sources of improvement in the quality of disease predictions. Hereby, an emphasis is on the evaluation of predictions for multiple diseases with simultaneous incorporation of their interdependence as an explanatory factor for quality improvement. This thesis:

1. detects general sources for improvement in predictive quality that emerge from mandatory preprocessing steps of survey data.
2. introduces a multi-label machine learning framework to compare the performance of algorithms and approaches that use interdependence between multiple disease targets.
3. identifies and discusses further factors that affect the predictive performance.

According to Grant et al. (2018), calibration and discrimination are the two main aspects that define the quality of predicted probabilities. The former assess the interpretability of predictions as probabilities, whereas the latter ensures a distinction between diseased and unaffected individuals. While recalibration is possible after any initial prediction, discrimination is inherent to the underlying algorithm and data. It determines the usability. Hence, this thesis evaluates algorithms and approaches by the discriminative power in their predicted disease probabilities for individuals (Grant et al. (2018), p. 205 f.).

The rest of this thesis is structured as follows. Chapter 2 introduces the theoretical framework underlying prediction with probabilistic algorithms based on a data sample. It is further extended to a multi-label framework with multiple diseases being of interest. This is followed by a discussion of candidate prediction algorithms in chapter 3. Based on the considered algorithms, chapter 4 introduces several candidate multi-label approaches for the prediction of multiple diseases with a simultaneous incorporation of their interdependence. The methodology and computational steps are lined out in chapter 5. Here, the German Health Update structure and the survey data are introduced in more detail. Mandatory preprocessing steps of the survey data are themed and conducted. Relevant performance metrics for the comparison of algorithms and multi-label approaches are considered. Algorithm optimization and the experimental setup are further clarified. The evaluation and discussion of the obtained results is conducted in chapter 6. Lastly, chapter 7 draws a conclusion and gives an outlook on possible further research in this field.

2 Theoretical Foundations

This chapter establishes the theoretical framework on which the probabilistic base algorithms from chapter 3 and the multi-label approaches from chapter 4 are based. Further, it is supposed to make apparent where the methodology as well as the experimental setup of chapter 5 stem from and provides the respective justification. Starting from a probabilistic perspective on binary classification, pertinent topics in and around statistical learning of probabilistic algorithms by the means of empirical risk minimization are introduced. Subsequently, the multi-label framework and its peculiarities are formulated in a probabilistic sense and as extension to the prediction of a single disease target.

2.1 Probabilistic Classifiers

Let $\mathcal{D} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$ be a dataset that consists of n instance pairs $(\mathbf{x}_i, y_i)$ with index $i \in \{1, \dots, n\}$. Those stem from random variables, thus y_i is a realization of a binary random variable $Y \in \{0, 1\}$ and $\mathbf{x}_i = [x_{i1} \ x_{i2} \ \dots \ x_{ik}]$ are realizations of a random vector $X = [X_1 \ X_2 \ \dots \ X_k] \in \mathcal{X}$ containing k random variables. Here, a realization x_{ij} with $j \in \{1, \dots, k\}$ is either real-valued or categorical by the input space $\mathcal{X}$. $\mathcal{D}$ consists of an $(n \times k)$ dimensional input matrix $\mathbf{X}$ with column vectors $\mathbf{X}_j = [x_{1j} \ x_{2j} \ \dots \ x_{nj}]^T$ and an $(n \times 1)$ dimensional target vector. It is assumed that $\mathcal{D}$ is generated by n independent and identically distributed draws from the unknown joint probability distribution $P(X, Y)$ on $\mathcal{X} \times \{0, 1\}$ (Ng & Jordan (2001), p. 2).

Further, assuming a sufficient association between the explanatory variables X and the target variable Y , a probabilistic classifier is in general defined by a function that maps $\mathcal{X} \rightarrow \{0, 1\}$, that relies on the prediction of an underlying probability distribution over the two classes of Y by conditioning on X . This implies that the classification takes place via the determination of the conditional probability $P(Y = y|X = \mathbf{x})$, followed by the assignment of the class that has the maximal probability. Note that $X = \mathbf{x}$ and $Y = y$ represents, with $\mathbf{x} \in \mathcal{X}$ and $y \in \{0, 1\}$, an arbitrary of all possible realization pairs in the following to simplify notation. Using the bayes theorem, a probabilistic classifier becomes

$$\arg \max_{y \in \{0, 1\}} P(y|\mathbf{x}) = \arg \max_{y \in \{0, 1\}} \frac{P(\mathbf{x}|y)P(y)}{P(\mathbf{x})} \propto \arg \max_{y \in \{0, 1\}} P(\mathbf{x}|y)P(y) \quad (2.1)$$

which yields the form of a maximum a posteriori estimator (Garg & Roth (2001), p. 181).

One can rewrite $P(\mathbf{x}|y)P(y) = P(\mathbf{x}, y)$ which is the joint probability distribution. Ng & Jordan (2001) point out that two possible approaches of predicting the conditional probability become apparent, namely via discriminative or generative classifiers. The former classifier models the posterior in equation 2.1 directly by learning patterns and decision boundaries to map $\mathbf{x}$ to y via the identification of $P(y|\mathbf{x})$. The latter models the joint probability $P(\mathbf{x}, y)$ and uses the bayes theorem to determine the posterior. Ng & Jordan (2001) provide evidence that discriminative classifiers outperform generative ones in classification tasks by comparing the respective standard classifiers. Further, the evidence for this result rises with the size of the underlying dataset used for discriminative classifier training. Moreover, the model assumptions of generative classifiers are stronger since the probability distribution over all $\mathbf{x}$ has to be identified. For this purpose, conditional independence between the explanatory variables is assumed. In the case of discriminative classifiers, assumption about $\mathbf{x}$ are not needed. Accordingly, only probabilistic discriminative classifiers are considered in the following, as they require weaker assumptions and may offer better performance given the dataset size specified in section 5.2.

Not every probabilistic output of discriminative classifiers can be directly interpreted as probability. This interpretation assumes that any probabilistic prediction $\hat{P}(y = 1|\mathbf{x})$ of 20% is consistent with the empirical observation that the positive class is found in 20% of the cases that have the same conditional probability. More generally, a probabilistic prediction is considered a score that is evaluated by scoring rules in the first place (Gneiting & Raftery (2007), p. 359). A rescaling is eventually needed to match an interpretable probability distribution. This process is called calibration and can be carried out after any score prediction. Niculescu-Mizil & Caruana (2005) show this for several algorithms introduced in chapter 3 using Platt Scaling and Isotonic Regression. Although this thesis focuses on improvements of the initial probabilistic score prediction, this motivates the use of a suiting optimization criterion that aims for probabilistic predictions. The approximation of the underlying true probability distribution is in expectation attained with a strictly proper scoring rule (Gneiting & Raftery (2007), p. 360). Proper implies an optimum at the true probability distribution and strict ensures its uniqueness. From optimizing any strictly proper scoring rule follows the minimization of an associated divergence between the probabilistic predictions and the true probability distribution (Gneiting & Raftery (2007), p. 361). In order to obtain predictions with probabilistic behavior from data, the framework of empirical risk minimization is introduced together with a strictly proper scoring rule in section 2.2. With that, all predicted scores are interpreted as expected and conditional probabilities.

$P(y = 1|\mathbf{x}) = 1 - P(y = 0|\mathbf{x})$ holds for the binary class case, where one yields the conditional probability of one class by determining the respective other. This makes the focus on probabilistic predictions for the class of primary interest sufficient. Let this be the

positive class and let $f : \mathcal{X} \rightarrow [0, 1]$, in line with the former paragraph, be a function that outputs the corresponding conditional probability by

$$f(\mathbf{x}) = P(y = 1|\mathbf{x}) . \quad (2.2)$$

Note that equation 2.1 dichotomizes the output. In contrast, equation 2.2 can be interpreted as a probabilistic soft decision result for the positive class. The function f is more generally called an algorithm. Candidates are introduced in chapter 3. If the algorithm is fitted to data, it is called a model $\hat{f}$.

2.2 Empirical Risk Minimization

In order to improve the quality of the probabilistic output of an algorithm on data, an appropriate loss function $\mathcal{L}(y, f(\mathbf{x}))$ is used. For probabilistic predictions, this implies a strictly proper scoring rule. Accounting for the relative uncertainty between classes, the desired behavior can be attained by considering the Binary Cross-Entropy (BC) loss

$$\mathcal{L}_{BC}(y, f(\mathbf{x})) = -y \log(f(\mathbf{x})) - (1 - y) \log(1 - f(\mathbf{x})) \quad (2.3)$$

that compares the true target class with the logarithm of the respective algorithm output (Gneiting & Raftery (2007), p. 365). The logarithm penalizes large differences which accounts for uncertainty. Following the maximum-likelihood principle, equation 2.3 coincides with the negative log-likelihood for binary classification (Bishop & Nasrabadi (2006), p. 209 f.). Simultaneously, it minimizes the Kullback-Leibler divergence between the true and expected probability distribution (Gneiting & Raftery (2007), p. 361 ff.). Minimizing the loss in equation 2.3 implies a numerical optimization of the associated likelihood and divergence.

$$R(f) = E[\mathcal{L}_{BC}(y, f(\mathbf{x}))] = \int \mathcal{L}_{BC}(y, f(\mathbf{x})) dP(X, Y) \quad (2.4)$$

is the so-called risk, the expectation of the loss function (Vapnik (1991), p. 832). According to equation 2.4, an optimal algorithm can in theory be identified directly when $P(X, Y)$ is known. However, this is generally not the case and an alternative, that does not rely on the joint probability distribution, is given by an approximation of $R(f)$ based on the dataset $\mathcal{D}$. This is called empirical risk minimization and can be written as

$$R_{emp}(f) = \frac{1}{n} \sum_{i=1}^n \mathcal{L}_{BC}(y_i, f(\mathbf{x}_i|\theta)) \propto \sum_{i=1}^n \mathcal{L}_{BC}(y_i, f(\mathbf{x}_i|\theta)) \quad (2.5)$$

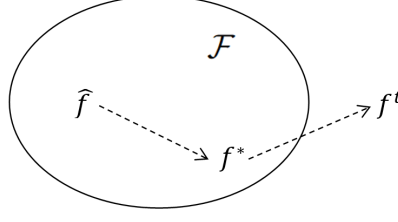


Figure 2.1: Schematic illustration of functions of interest (Nam (2019), p. 9)

where θ denotes the vector of parameter when f is parameterized (Vapnik (1991), p. 832). The empirical risk in equation 2.5 is reduced when the output of the algorithm improves. Here, the loss is also refereed to as objective function. The approximation can be considered good if $\mathcal{D}$ is an unbiased sample from $P(X, Y)$. The empirical risk is a random variable that converges in probability to the population risk, as shown in equation 2.6. This follows from the law of large numbers (Niyogi & Girosi (1996), p. 4). $\forall \epsilon > 0$ holds

$$P(|R(f) - R_{emp}(f)| < \epsilon) = 1 \quad \text{as} \quad n \rightarrow \infty \quad \text{and} \quad R_{emp}(f) \xrightarrow{p} R(f). \quad (2.6)$$

Let $\mathcal{F}$ be a hypothesis space that restricts the functional forms of f and Θ be the parameter space of the respective function f . Following Niyogi & Girosi (1996), this implies that the objective for the learning process of the function in $R_{emp}(f)$ is more generally given by

$$f^* = \arg \min_{f \in \mathcal{F}} R_{emp}(f) \quad \text{which implies} \quad \theta^* = \arg \min_{\theta \in \Theta} R_{emp}(\theta) \quad (2.7)$$

if f is parameterized. Equation 2.7 shows that a suiting functional form and eventually its parameters have to be determined in order to minimize the empirical risk. Nevertheless, it is not guaranteed that $\hat{f}$ coincides with the true population model f^t which can, for example, lie outside of $\mathcal{F}$. Even on $\mathcal{F}$, the risk minimal model f^* is hard to identify. This can occur when f^* is infeasible due to the computational complexity of numerical optimization and identification that can suffer from the curse of dimensionality in the functional form. In line with equation 2.6 and a finite sample, the deviations are shown in Figure 2.1.

Further conditions for model identification are required in order to train a suiting candidate for $\hat{f}$. Note that the main benefit of training $\hat{f}$ arises from predictions of an unknown target y with new instances of $\mathbf{x}$. Here, the model is required to generalize to non-training data. This implies that the objective of finding a more general model $\hat{f}$ close to f^t rather than optimizing the model towards f^* , i.e. to specific data structures. The deviation that is to be minimized is called Generalization Error (GE) of $\hat{f}$. It decomposes in two possible error sources (Niyogi & Girosi (1996), p. 5 f.). If one assumes $f^t \in \mathcal{F}$, then those errors occur because of the deviation of $R_{emp}(f^*)$ from $R(f^t)$ due to the finite dataset $\mathcal{D}$ and from $R_{emp}(f^*)$

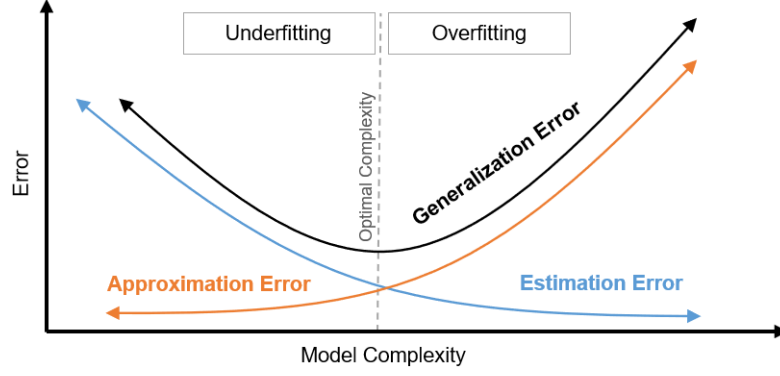


Figure 2.2: Overfitting and generalization error decomposition by complexity of $\hat{f}$ stemming from over-parametrization and irrelevant variables (James et al. (2013), p. 36)

being unidentifiable. The former term on the right hand side of equation 2.8 represents the approximation error and the latter term is referred to as estimation error. The approximation error describes how close the empirical risk of the optimal model $R(f^*)$ is to the true risk $R(f^t)$ because of $\mathcal{D}$ being a finite sample. The estimation error relies only on the estimation capability of $\hat{f}$ that indicate how close the empirical risk of the identified model is to the optimal empirical risk model $R(f^*)$ on $\mathcal{F}$ given $\mathcal{D}$. Considering a high level of functional complexity in the design of $\hat{f}$ potentially reduces the estimation error at the expense of an increase in the approximation error, which is called overfitting (Hastie et al. (2009), p.364 and p.398). Note that it is induced by over-parameterization in hyperparameters and an consideration of irrelevant variables in X . As stated in Barron (1994) more formally, this can be seen in the decomposition of the GE between the true risk and the empirical risk minimum into approximation error and estimation error. The triangle inequality yields

$$GE = \|R(f^t) - R_{emp}(\hat{f})\| \leq \|R(f^t) - R(f^*)\| + \|(R(f^*) - R_{emp}(\hat{f}))\|. \quad (2.8)$$

A high GE harms the predictive performance of $\hat{f}$ and therefore has to be considered in the following analyses. The emerging trade-off is called the bias-variance trade-off, which can be stated equivalently to equation 2.8 based on the decomposition of the mean squared error in bias and variance (James et al. (2013), p. 34). It is illustrated in Figure 2.2. It is of interest to identify $\hat{f}$ based on $\mathcal{D}$ such that the GE is minimized in terms of its two error sources.

In order to obtain an estimate for the GE on $\mathcal{D}$, resampling is used. A random split of the dataset that preserves the dataset distributions in a training set D_{train} and a test set D_{test} is utilized. As suggest by the set names, D_{train} is used to identify the model $\hat{f}$. A loss calculated on the known data, i.e. instances the model is trained on, represents an inner loss. Subsequently, the unseen data D_{test} is treated as if the target realizations were unknown. D_{test} is used for the prediction of conditional probabilities and their evaluation against the

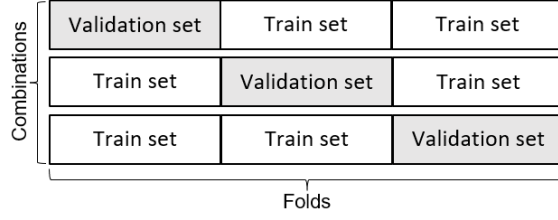


Figure 2.3: Example splits of a 3-fold cross-validation

true outcomes. Consequently, an estimator for the GE between D_{train} and D_{test} is given by

$$\widehat{GE} = \frac{1}{|D_{test}|} \sum_{(x_i, y_i) \in D_{test}} \mathcal{L}_{BC}(y_i, \hat{f}_{D_{train}}(x_i)) \quad (2.9)$$

where $|D_{test}|$ is the set cardinality. The loss that is estimated in equation 2.9 is called outer loss. It can be substituted with a wide range of other metrics than the binary cross-entropy loss of equation 2.3 to assess model performance. Accordingly, a performance metric is a measure that can be chosen to evaluate the problem at hand, see section 6.1. In that sense, $\widehat{GE}$ can be used for comparisons of predictive performance for different forms of optimized algorithms $\hat{f}$. Identifying a model on D_{train} that greedily minimizes the empirical risk via the inner loss does not imply an optimization of $\widehat{GE}$, but rather leads to overfitting. As discussed in Mukherjee et al. (2006), a model validation step has to be integrated in the identification process to obtain a generalizing model. This is done by resampling of D_{train} into partitions such that the data is divided in M equally sized and non overlapping folds. Let one fold denote a validation set $\mathcal{D}_{val} \subset \mathcal{D}_{train}$ with M possible choices, then there exists a set with all combinations of $\mathcal{D}_{val}^{[m]}$ and $\mathcal{D}_{train}^{[m]} = \mathcal{D}_{train} \setminus \mathcal{D}_{val}^{[m]}$ with iterator $m \in (1, \dots, M)$. An expected GE

$$E[\widehat{GE}] = \frac{1}{M} \sum_{m=1}^M \frac{1}{|\mathcal{D}_{val}^{[m]}|} \sum_{(x_i, y_i) \in \mathcal{D}_{val}^{[m]}} \mathcal{L}_{BC}(y_i, \hat{f}_{\mathcal{D}_{train}^{[m]}}(x_i)) \quad (2.10)$$

can be determined. This is illustrated in Figure 2.3. Note that the data of the respective model is not used for validation in equation 2.10. The so-called outer or validation loss for $\mathcal{D}_{train}$ is obtained via evaluations across all possible validation sets given by the M folds. A model $\hat{f}$ that minimizes the expected GE instead of the inner loss is intended to minimize the GE of equation 2.10. The inner loss is neither utilized in the model identification via equation 2.10, nor during evaluation by equation 2.9. If one chooses a initial split according to equation 2.9 and identifies optimized models via validation according to equation 2.10, a simple framework for the identification of generalizing models and comparison of different algorithm types is generated. Note that this setup directly applies to the identification and evaluation of the probabilistic base algorithms in chapter 3. For the evaluation of

the multi-label approaches of chapter 4, a more sophisticated framework is introduced in section 5.4. However, a formal extension to the multi-label framework and a discussion of its peculiarities is required first.

2.3 Multi-label Extensions

Only one binary target variable is of interest in section 2.1. Due to the categorical dichotomy, the realization of the variable can be interpreted as a so-called label. The target realizations determine whether some instance i shows the characteristic underlying the binary variable. Its occurrence is indicated by the positive class. Let the focus now be on the positive class labels of L binary target variables. The random vector of binary random variables $[Y_1 Y_2 \dots Y_L] \in \mathcal{Y}$ with $\mathcal{Y} \in \{0, 1\}^L$ and realizations $\mathbf{y}_i = [y_{i1} y_{i2} \dots y_{iL}]$ induces the multi-label framework. $\mathcal{D}$ now consists of n instances $(\mathbf{x}_i, \mathbf{y}_i)$ that are assumed to be independent and identically distributed draws from the unknown joint probability distribution on $\mathcal{X} \times \mathcal{Y}$. Keeping the input matrix equal to section 2.1, the target matrix $\mathbf{Y}$ with column vectors $\mathbf{Y}_l = [y_{1l} y_{2l} \dots y_{nl}]^T$ and $l \in \{1, 2, \dots, L\}$ becomes $(n \times L)$ dimensional. Any data split in training, test or validation set leads to a equivalent split in the the instances of the input and target matrix, e.g. from $\mathcal{D}_{train}$ and $\mathcal{D}_{test}$ follows $\mathbf{X}_{train}$ and $\mathbf{X}_{test}$ as well as $\mathbf{Y}_{train}$ and $\mathbf{Y}_{test}$. For the sake of notation, let $\mathbf{x}$ and $\mathbf{y}$ denote any pair of random vector realizations among all possible realizations.

Taking a probabilistic point of view, one can directly rewrite the first equality in equation 2.1 by substituting the random realization y with random realizations $\mathbf{y} \in \{0, 1\}^L$ from $\mathcal{Y}$ in order to obtain a probabilistic multi-label classifier. Basing this paragraph on Dembczyński et al. (2012), the objective is to obtain the optimal predictions of $P(y_l = 1|\mathbf{x}) \forall l$ with labels $l \in \{1, 2, \dots, L\}$, which are the so-called marginal distributions. In order to attain this, it is of further interest to utilize, apart from $\mathbf{x}$, any dependency between the target variables such that the predictive performance is improved by empirical risk minimization, i.e. for a loss $\mathcal{L}$ of interest. For that, the conditional joint distribution of the binary target variables $P(\mathbf{y}|\mathbf{x})$, which is already conditioned on $\mathbf{x}$, becomes of relevance. If one assumes conditional independence between the L target labels, the conditional joint distribution factorizes to

$$P(y_1, y_2, \dots, y_L|\mathbf{x}) = \prod_{l=1}^L P(y_l|\mathbf{x}) . \quad (2.11)$$

It follows that the training of L independent models is sufficient if equation 2.11 hold. Only the marginal distributions have to be assessed according to section 2.2. However, utilizing existing conditional dependencies between the L target variables increases the difficulty. In

this case, the product rule theoretically implies that

$$P(y_1, y_2, \dots, y_M | \mathbf{x}) = P(y_1 | \mathbf{x}) \prod_{l=2}^L P(y_l | y_1, \dots, y_{l-1}, \mathbf{x}) \quad (2.12)$$

decomposes the conditional joint distribution. Any candidate multi-label approach, as a function $h : \mathcal{X} \rightarrow [0, 1]^L$, is expected to be able to capture and take advantage of the dependencies across the labels. With the positive class of primary relevance, an extension from section 2.1 to the multi-label framework follows from h taking the vector form

$$h(\mathbf{x}) = [h_1(\mathbf{x}) \ h_2(\mathbf{x}) \ \dots \ h_L(\mathbf{x})] \quad \text{where} \quad h_l(\mathbf{x}) = P(y_l = 1 | Dep(y_l), \mathbf{x}) \quad (2.13)$$

as output (Dembczyński et al. (2010), p. 2). Equation 2.13 conditions on both $\mathbf{x}$ and $Dep(y_l)$. The letter $Dep(y_l)$, i.e. without revealing how, represents all other target variables that y_l depends on (Fu (2018), p. 16 f.). Note that $Dep(y_l)$, in contrast to equation 2.12, highlights that the structure of the dependencies are in practical applications often unknown. From a theoretic perspective, equation 2.13 can be approached by utilizing the dependencies between the marginal distributions or alternatively by estimating $P(\mathbf{y} | \mathbf{x})$ followed by a reasonable marginalization procedure (Dembczyński et al. (2012), p. 14). This corresponds to the approach separation in Problem Transformation and Algorithm Adaptation of chapter 4. However, assuming conditional independence $Dep(y_l)$ vanishes from equation 2.13 and equation 2.12 simplifies to equation 2.11. How $Dep(y_l)$ is accounted for in h , such that the incorporated dependencies improve the predictions $h_l(\mathbf{x})$ for each label l , is a major aspect and in the spotlight in chapter 4.

The identification and evaluation of $h(\mathbf{x})$ are again conducted in the framework of empirical risk minimization. The BC loss $\mathcal{L}_{BC}$ from equation 2.3 becomes a multi-label version for L labels by aggregation. Note that the decomposeability of the loss as well as the evaluation in section 5.3 ensures a label-wise assessment as well. This is in line with the objective of optimizing $P(y_l = 1 | \mathbf{x}) \ \forall l$ using $Dep(y_l)$ (Dembczyński et al. (2012), p. 13). Instead of $\mathcal{L}_{BC}$, one can define the Multi-label Binary Cross-Entropy (MBC) loss

$$\mathcal{L}_{MBC}(\mathbf{y}, h(\mathbf{x})) \propto - \sum_{l=1}^L y_l \log(h_l(\mathbf{x})) - (1 - y_l) \log(1 - h_l(\mathbf{x})) , \quad (2.14)$$

which can be minimized up to a normalizing constant by proportionality. Let $\mathcal{H}$ be a hypothesis space that restricts the functional forms of h and Θ be the parameter space of the respective function h . Accordingly, the model that minimizes the empirical risk is given by

the optimization of

$$h^*(\mathbf{x}) \propto \arg \min_{h \in \mathcal{H}} \sum_{i=1}^n \mathcal{L}_{MBC}(\mathbf{y}_i, h(\mathbf{x}_i)) \quad \text{or} \quad h^*(\theta) \propto \arg \min_{\theta \in \Theta} \sum_{i=1}^n \mathcal{L}_{MBC}(\mathbf{y}_i, h(\mathbf{x}_i|\theta)) \quad (2.15)$$

if h is parameterized. Note that the concepts of generalization and validation via resampling directly carry over to the multi-label framework from section 2.2. Having a random split into D_{train} and D_{test} , one can identify an optimized model with equation 2.15 via outer loss minimization by cross-validation. This implies identification of $\hat{h}$ in

$$E[\widehat{GE}_{ML}] = \frac{1}{M} \sum_{m=1}^M \frac{1}{|\mathcal{D}_{val}^{[m]}|} \sum_{(\mathbf{x}_i, \mathbf{y}_i) \in \mathcal{D}_{val}^{[m]}} \mathcal{L}_{MBC}(\mathbf{y}_i, \hat{h}_{\mathcal{D}_{train}^{[m]}}(\mathbf{x}_i)) \quad (2.16)$$

using M folds such that the loss from equation 2.14 is minimized. With the optimized model fitted to D_{train} , a GE estimate follows from evaluation based on D_{test} via

$$\widehat{GE}_{ML} = \frac{1}{|D_{test}|} \sum_{(\mathbf{x}_i, \mathbf{y}_i) \in D_{test}} \mathcal{L}_{MBC}(\mathbf{y}_i, \hat{h}_{D_{train}}(\mathbf{x}_i)) . \quad (2.17)$$

More generally, the equations 2.16 and 2.17 are used in section 5.4 to optimize algorithms and approaches as well as to generate experimental setups in which the predictive performance of different multi-label approaches and underlying probabilistic base algorithms are compared. The probabilistic base algorithms are introduced in chapter 3 followed by the multi-label approaches in chapter 4. In order to evaluate the predictive performance further, several performance metrics can be considered instead of the MBC in equation 2.17 to assess the properties of predictions. Candidates of interest are discussed in section 6.1.

3 Probabilistic Base Algorithms

Section 2.3 show that multi-label approaches are linked to the prediction of the marginal distributions $P(y_l = 1 | Dep(y_l), \mathbf{x})$, which can be done using a probabilistic algorithm f as introduced in section 2.1. Accordingly, the performance of multi-label approaches depends on the underlying base algorithm in section 4.1. As pointed out in section 4.2, base algorithms can be adapted to model the multi-label problem directly and hence provide a theoretical foundation. Candidate algorithms f are introduced in this chapter. The emphasis is on the generation of conditional probabilities as well as central hyperparameters that can be optimized in order to obtain an improvement in terms of empirical risk minimization.

An overview of potential candidates with relation to the problem at hand are given in the comparative study of Uddin et al. (2019). In section 2.1, generative methods got ruled out. However, some discriminative algorithms are still not well suited because of several data peculiarities. The survey data consist almost exclusively of categorical variables with numerous category levels. The encoding described in subsection 5.2.4 increases the dimensionality and induces noise in the case of low frequency category levels. Distance-based algorithms are faced with problems in the pattern identification. More specifically, Support Vector Machines as introduced by Boser et al. (1992) do not scale well to large datasets and k-Nearest-Neighbours stemming from Altman (1992) rely on calculation of high-dimensional distances. Even though several steps were undertaken to account for this problems, none ensured a competitive applicability of distance-based algorithms to the problem at hand. Apart from that, a direct output of conditional probabilities is not possible for both. For this reasons distance-based algorithms are disregarded.

Accordingly, the focus is on a regularized version of Logistic Regression, Decision Trees in Bagging and Boosting ensembles and Feed-forward Neural Networks as algorithm candidates for the function f . Those are introduced in line with their implementation and as practically used in chapter 5. Nevertheless, the focus is on the illustration of the functioning and theoretic benefits of the respective algorithm. The idea of the underlying numerical solving method as well as efficiency and optimization aspects are only briefly sketched and then referenced.

3.1 Regularized Logistic Regression

If the explanatory variables are sufficiently associated with the target variable, then the regularized Logistic Regression (LR) is a frequently used algorithm to model this the relationship between $\mathbf{x}$ and y . It is based on the linear regression of the form

$$y = \beta_0 + \mathbf{x}\tilde{\beta} + \varepsilon \quad \text{with} \quad \varepsilon \sim N(0, \sigma^2) \quad \text{and} \quad \sigma^2 > 0, \quad (3.1)$$

where $\beta = [\beta_0 \ \beta_1 \ \dots \ \beta_k]^T \in \mathbb{R}^p$ is a vector of parameters with $p \in \{0, 1, \dots, k\}$ that have to be estimated (Hastie et al. (2009), p. 44 ff.). β_0 resembles the intercept and $\tilde{\beta} = [\beta_1 \ \dots \ \beta_k]^T \in \mathbb{R}^k$ are coefficients that equation 3.1 is linear in. ε denotes an error term that is included to capture unsystematic deviations from the binary target y and the systematic part explained by $X\beta$. This error term is minimized in the estimation of the parameters β which then, by suppressing the intercept in the notation, yields the residuals $e_i = y_i - \mathbf{x}_i\hat{\beta}$.

Further, Tibshirani (1996) introduced a complexity penalty λ as a hyperparameter that is imposed on $\tilde{\beta}$ and shrinks the coefficients. The L_1 -norm is used as constraint, which is driven by practical relevance in chapter 5. The parameter estimation from equation 3.1 becomes

$$\arg \min_{\beta \in \mathbb{R}^p} \left\{ \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^k x_{ij}\beta_j \right)^2 + \lambda \|\tilde{\beta}\|_1 \right\} = \arg \min_{\beta \in \mathbb{R}^p} \left\{ R_{emp}(\beta) + \lambda J(\tilde{\beta}) \right\}. \quad (3.2)$$

The first term in equation 3.2 stems from minimizing the mean squared error loss of e underlying equation 3.1 in the identification of β and the latter resembles the constraint J with the penalty. The constraint can be rewritten into $\|\tilde{\beta}\|_1 = \sum_{j=1}^k |\beta_j| \leq t$ where $\tilde{\beta}$ are the restricted coefficients (Hastie et al. (2009), p. 68). Figure 3.1 provides insights into the coefficient shrinking in $\mathbb{R}^2$, i.e. in the case of the constraint $|\beta_1| + |\beta_2| \leq t$. Minimizing equation 3.1 alone results in the unrestricted coefficients $\hat{\beta}^U$. In Figure 3.1, the permissible area defined by the constraint is marked in light blue and the unrestricted solution $\hat{\beta}^U$ is not part of it. The orange level lines resemble possible solutions of the objective function in $R_{emp}(\beta)$ originating from equation 3.1. Those show that the geometric form of the permissible solution area given by the L_1 penalty yields a corner solution for the optimization problem underlying $\hat{\beta}$. This implies in Figure 3.1 that β_1 shrinks and β_2 is set to 0. The larger t is, the less likely a corner solution becomes. For a sufficiently large t , $\hat{\beta}^U$ is within the permissible area. The number of vertices of the permissible area grows exponentially in the dimension k and thus the probability for coefficients with the value zero increases as well (Hastie et al. (2015), p. 12). Implicitly, a subset selection is performed by equation 3.2 via the determination of coefficients with respect to the L_1 -norm penalty.

The L_1 -norm is not differentiable in any coordinate origin, thus a closed form solution does not exist. However, one can consider an orthonormal basis of the $(n \times k)$ input matrix with

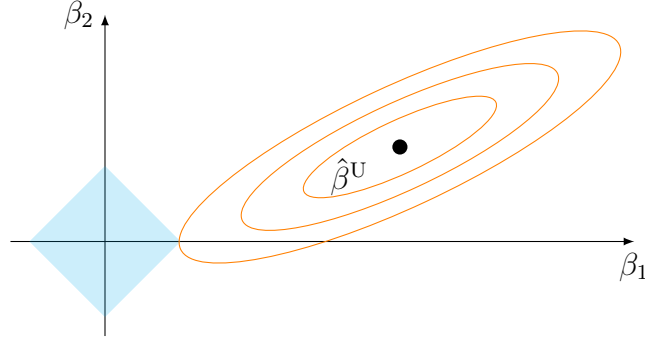


Figure 3.1: Visualization of coefficient shrinking by the L_1 -norm (blue) and the level lines of the unrestricted objective function (orange) (Hastie et al. (2009), p. 71)

$\mathbf{X}^T \mathbf{X} = \mathbf{I}_k$ for a theoretic result (Tibshirani (1996), p. 285). According to convex analysis, a sub-gradient is considered (Hastie et al. (2015), p. 99 f.). Optimizing equation 3.2 becomes

$$\nabla_{\tilde{\beta}} R_{reg}(\tilde{\beta}) = \nabla_{\tilde{\beta}} R_{emp}(\tilde{\beta}) + \nabla_{\tilde{\beta}} \lambda J(\tilde{\beta}) = 2 \sum_{i=1}^n (y_i - \mathbf{x}_i^T \tilde{\beta}) (-\mathbf{x}_i^T) + \lambda s(\tilde{\beta}) \stackrel{!}{=} 0, \quad (3.3)$$

were the intercept can be omitted. Note that the solution for the unrestricted coefficients is $\hat{\beta}^U = (\sum_{i=1}^n \mathbf{x}_i^T \mathbf{x}_i)^{-1} \sum_{i=1}^n \mathbf{x}_i^T y_i$. Further, $\forall j \in \{1, \dots, k\}$ the sub-gradient $s(\tilde{\beta})$ is the respective sign of $\tilde{\beta}_j$ if $\tilde{\beta}_j \neq 0$ (Hastie et al. (2015), p. 9). The problem in equation 3.2 is separable and hence allows a coordinate-wise optimization of each β_j (Tibshirani (1996), p. 269). It results in the so-called soft-thresholding operator

$$\hat{\beta}_j = s(\hat{\beta}_j^U) \max\left(|\hat{\beta}_j^U| - \frac{\lambda}{H_{jj}}, 0\right) = \begin{cases} \hat{\beta}_j^U + \frac{\lambda}{H_{jj}}, & \text{if } \hat{\beta}_j^U < -\frac{\lambda}{H_{jj}} \\ 0, & \text{if } \hat{\beta}_j^U \in [-\frac{\lambda}{H_{jj}}, +\frac{\lambda}{H_{jj}}] \\ \hat{\beta}_j^U - \frac{\lambda}{H_{jj}}, & \text{if } \hat{\beta}_j^U > +\frac{\lambda}{H_{jj}} \end{cases} \quad (3.4)$$

with H_{jj} as the diagonal of the Hessian matrix (Hastie et al. (2015), p. 15). Every $\hat{\beta}_j^U$ gets distorted towards 0 in equation 3.4 and is set to 0 for $\hat{\beta}_j^U \in [-\frac{\lambda}{H_{jj}}, \frac{\lambda}{H_{jj}}]$. In general, the penalty is scale variant and therefore relies on normalized data, which also allows to drop the intercept. The regularization of LR can be seen as a continuous model selection process, that eventually excludes variables from the estimation and reduced the complexity via λ (Hastie et al. (2009), p. 69 f.). Recalling Figure 2.2, this can improve the predictive performance and mitigate overfitting if the estimation error is increased for an over-proportional decrease in the approximation error.

However, estimates of β from equation 3.2 are not directly suited to represent an algorithm f . $\beta_0 + \mathbf{x}\tilde{\beta}$ can take values outside of $[0, 1]$, hence a logit-transformation must be applied to

the linear predictor (Friedman et al. (2010), p. 8). This yields the conditional probability

$$f_{LR}(\mathbf{x}) = \frac{1}{1 + \exp(-(\beta_0 + \mathbf{x}\tilde{\beta}))} . \quad (3.5)$$

Still, there is no general closed-form solution. Inserting equation 3.5 into the BC loss from equation 2.3 and considering a L_1 -norm regularization, one can rewrite the optimization problem to

$$-\ell_{\mathcal{L}}(\beta) = -\frac{1}{n} \sum_{i=1}^n \left\{ y_i(\beta_0 + \mathbf{x}_i\tilde{\beta}) - \log(1 + \exp(\beta_0 + \mathbf{x}_i\tilde{\beta})) \right\} + \lambda \|\tilde{\beta}\|_1 \quad (3.6)$$

(Friedman et al. (2010), p. 8). The penalized negative log-likelihood in equation 3.6 is convex and the likelihood part differentiable, hence the problem can be solved by iterative optimization that identifies $\hat{\beta}$ (Hastie et al. (2015), p. 35). Here, Friedman et al. (2010) use the second order Taylor expansion of the likelihood part with the current iteration estimates that are given by $(\dot{\beta}_0, \dot{\beta})$. The resulting quadratic approximation is

$$\ell_Q(\beta) = -\frac{1}{2n} \sum_{i=1}^n w_i (z_i - \beta_0 + \mathbf{x}_i\tilde{\beta})^2 + C(\dot{\beta}_0, \dot{\beta})^2 . \quad (3.7)$$

Equation 3.7 resembles an iteratively weighted least square problem with a constant $C(\dot{\beta}_0, \dot{\beta})^2$, weights $w_i = \dot{f}(\mathbf{x}_i)(1 - \dot{f}(\mathbf{x}_i))$ and the current target $z_i = \dot{\beta}_0 + \mathbf{x}_i\dot{\beta} + \frac{y_i - \dot{f}(\mathbf{x}_i)}{w_i}$. $\dot{f}(\mathbf{x}_i) = f(\mathbf{x}_i|\dot{\beta}_0, \dot{\beta})$ is evaluated at the current iteration estimates. The minimization of the likelihood part is done by a local approximation. This implies that the underlying optimization problem can be stated by

$$\arg \min_{\beta \in \mathbb{R}^p} \left\{ \ell_Q(\beta) + \lambda \|\tilde{\beta}\|_1 \right\} = \arg \min_{\beta \in \mathbb{R}^{k+1}} R_{LR}(\beta_0, \tilde{\beta}) . \quad (3.8)$$

Equation 3.8 is a penalized weighted least-squares problem, that can, like equation 3.2, be solved by coordinate descent (Hastie et al. (2015), p. 112 ff.). This implies that β_j is initialized and then iteratively updated, e.g. by cycling j through $\{1, 2, \dots, k\}$ until convergence. This is done by minimizing the partial residuals $e_j \forall j$. For that, the derivative $\frac{\partial R_{LR}(\beta_0, \tilde{\beta})}{\partial \beta_j}$ is evaluated at the current iteration estimates $(\dot{\beta}_0, \dot{\beta})$ (Friedman et al. (2010), p. 5 ff.). An improved and more efficient numerical solving technique is given by Yuan et al. (2012) which is the current basis of the LR implementation of Fan et al. (2008) that is used in chapter 5.

3.2 Decision Tree Ensembles

Classification Trees (CT) are the non-parametric equivalent of a LR. Breiman et al. (2017) introduces a frequently used implementation. CT are trained by recursive binary partitioning of the dataset $\mathcal{D}$ (Hastie et al. (2009), p. 305). With dataset cardinality $|\mathcal{D}|$ and column vectors $\mathbf{X}_j$ of all explanatory variable realizations, one split point s is identified by exhaustive search such that the split in column j maximizes the impurity reduction of the target realizations in the resulting two subsets. For the initial split, those are $\mathcal{D}_1 = \{(\mathbf{x}_i, y_i) \in \mathcal{D} \mid \mathbf{x}_{ij} < s\}$ and $\mathcal{D}_2 = \mathcal{D} \setminus \mathcal{D}_1$. With the impurity measure I , the reduction in the target impurity becomes

$$\Delta I(j, s) = I(\mathcal{D}) - \left(\frac{|\mathcal{D}_1|}{|\mathcal{D}|} I(\mathcal{D}_1) + \frac{|\mathcal{D}_2|}{|\mathcal{D}|} I(\mathcal{D}_2) \right). \quad (3.9)$$

The impurity reduction in equation 3.9 originates from the weighted average of the subset impurities. A suitable impurity measure for probabilistic prediction is the Shannon Entropy

$$I(\mathcal{D}_1) = -\pi \log(\pi) - (1 - \pi) \log(1 - \pi) \quad \text{with} \quad \pi \in [0, 1] \quad (3.10)$$

which is the associated information measure of the BC in equation 2.3 (Gneiting & Raftery (2007), p. 364 f.). Note that π is the proportion of the positive class in the respective node in equation 3.9. Splitting is repeated for each subset until some stopping criterion or a pure class subset is reached. A stylized example is given in Figure 3.2. The initial split at the so-called root node is followed by splits in child nodes that terminate in the leafs. This represents a fitted model $\hat{f}$. Each node shows the variable and the respective split point that is used. The colored leafs in Figure 3.2 display top-down the majority class of the leaf, followed by the proportion of the positive class π as well as the data share that got allocated to the respective leaf. In fact, the probability output of the CT coincides with π in the leafs. If the splits allocate x to leaf q of Q tree leafs, the conditional probability follows by

$$f_{CT}(\mathbf{x}) = \frac{1}{n_q} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{D}_q} \mathbb{1}_{\{y_i=1\}}. \quad (3.11)$$

n_q in equation 3.11 is the number of training set instances in leaf q . An adjustment of CT to categorical input variables is intuitive. The categories are ordered according to their positive class proportion and splitted in the binary subgroups containing an equal number of categories (Hastie et al. (2009), p. 310). If this is not possible due to an uneven number of categories, an evaluation of Shannon Entropy identifies one of the two split candidates.

Note that unrestricted CT are grown in a greedy fashion which can result in deep and complex structures with leafs that are pure with respect to one class. However, this induces the overfitting problem from Figure 2.2. Further, a peculiarity of trees is their high structural

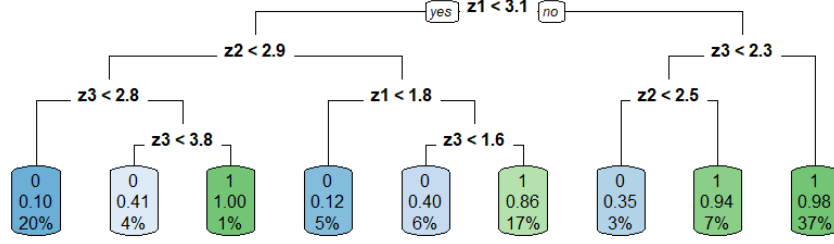


Figure 3.2: Classification Tree example on simulated data

instability with respect to changes in the dataset $\mathcal{D}$ and hence estimates for equation 3.11 vary (James et al. (2013), p. 340). In order to address overfitting, hyperparameters that reduce the tree depth can be considered. The instability can be mitigated or even used advantageously with respect to predictive performance by combining several trees into a homogeneous CT ensemble. The principles that are commonly used to build such ensembles are Bagging introduced by Breiman (1996) and Boosting as discussed in Schapire (1990). The former utilizes so-called weak learners that are homogeneous, i.e. of the same algorithm class, in parallel and aggregates their predictions. The latter builds weak learners sequentially and updates the predictions along the sequence. Those homogeneous ensemble methods are introduced in the following and detailed at the example of CT as weak learners.

3.2.1 Random Forest

Building a homogeneous ensemble by growing several CT in parallel introduces the Bagging principle. The name of the algorithm stems from bootstrap and aggregation (Breiman (1996), p.123). The idea is to draw M bootstrap samples, i.e. by sampling with replacement, from $\mathcal{D}$ and fit a weak learning algorithm to each sample $m = \{1, 2, \dots, M\}$. Here, the instability of the CT yield varying results with the changing data basis. In the case of probabilistic prediction, the final prediction follows from the aggregation of all weak learner predictions via averaging or majority voting. The contribution of each tree is given by the proportion of the positive class according to the leafs as seen in Figure 3.2. Bagging can be utilized as meta algorithm that increases the predictive performance by using CT as weak learner (Breiman (1996), p.124). However, the principle hold more generally and applies to any other weak learner given its instability.

An algorithm that optimizes the Bagging principle with respect to CT is introduced in Breiman (2001) and called Random Forest (RF). The main extension to Bagged CT is the higher level of randomization that is imposed on each tree. This emerges from two sources, namely from bootstrap samples that have less than $|\mathcal{D}|$ entries and from the restriction of the column vectors with explanatory variable realizations that are used for splits. The latter

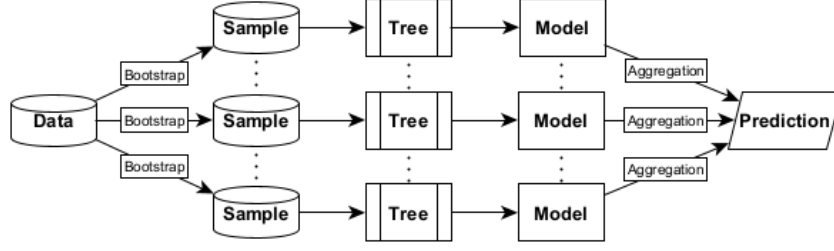


Figure 3.3: Conceptual illustration of Random Forest

follows from the selection of a random subset with less than k candidates and applies to every node of Figure 3.2. Performance improvements stem from an increased variability in each weak learner as well as the decorrelation of the weak learners (Breiman (2001), p.9 ff.). This implies that the individual weak learner errors differ from each other, whereby the information gain caused by aggregation is higher. Here, the implementation of RF used in chapter 5 utilizes averaging via

$$\frac{1}{M} \sum_{m=1}^M f_{CT}^{[m]}(\mathbf{x}) = f_{RF}(\mathbf{x}) \quad (3.12)$$

where M is the number of weak learners. An illustration of the conceptual steps of the RF algorithm is given in Figure 3.3. In equation 3.12, no parameters θ that have to be estimated in comparison to section 3.1. However, in line with the considerations for CT and Bagging, there exist several relevant hyperparameters. Firstly, M , the number of CT, can be increased. However, the benefits for predictive performance diminish with each additional tree. The level of variability in the CT can be controlled by the bootstrapped sample size, which is measured as proportion of the number of total entries n . Likewise the level of randomization changes with the number of randomly selected explanatory variables at each node. Finally, the complexity of the trees can be restricted and controlled by setting a maximum depth.

3.2.2 Gradient Boosted Trees

Gradient Boosting (GB) is a further ensemble principle that can be utilized to build homogeneous ensembles from weak learners. It relies on gradient descent applied to an objective function. The underlying loss function is required to be differentiable, convex and unconstrained. The steepest gradient step is identified for loss minimization and used to derive an update of the previous iteration. In contrast to Bagged CT in Figure 3.3, this results in a

sequence of updates via CT. One update of a boosting iteration is

$$g^{[m]}(\mathbf{x}) = g^{[m-1]}(\mathbf{x}) + b^{[m]}(\mathbf{x}) \quad \text{with} \quad b^{[m]}(\mathbf{x}) = \sum_{q=1}^{Q^{[m]}} c_q^{[m]} \mathbb{1}_{\{\mathbf{x} \in T_q^{[m]}\}} , \quad (3.13)$$

where $b^{[m]}(\mathbf{x})$ is a regression tree at iteration m , $T_q^{[m]}$ represents one of $Q^{[m]}$ leafs and $c_q^{[m]}$ is the average value in that leaf (Hastie et al. (2009), p. 364 and p. 307). In comparison to CT, regression trees differ in the splitting criterion used. The variance of the continuous target variable in the sub-samples of the initial split $\mathcal{D}_1$ and $\mathcal{D}_2$ and onwards is minimized. The regression tree $b^{[m]}(\mathbf{x})$ in equation 3.13 is fitted to the so-called pseudo residuals $e^{[m-1]}$ that resemble the error in the output of the previous sequence.

In fact, the steepest descent of the empirical risk gradient evaluated at the currently available sequence coincides with the pseudo residuals of interest (Hastie et al. (2009), p. 358 f.). This becomes

$$e^{[m-1]} = -\nabla R(g(\mathbf{x})) = -\frac{\delta R(g(\mathbf{x}))}{\delta g(\mathbf{x})} \bigg|_{g(\mathbf{x})=g^{[m-1]}(\mathbf{x})} . \quad (3.14)$$

According to equation 3.14, the pseudo residuals are given by a vector of partial derivatives, that in contrast to coordinate descent in section 3.1 incorporates the update of all k input dimensions. The loss function suggested in Friedman (2001) coincides with the objective function of LR, i.e. the unconstrained negative log-likelihood part in equation 3.6. Substituting the linear regression by any suiting regression $g(\mathbf{x})$ yields the Bernoulli loss which is more generally given by

$$L_B(y, g(\mathbf{x})) = -yg(\mathbf{x}) + \log(1 + \exp(g(\mathbf{x}))) . \quad (3.15)$$

The negative derivative of the objective function generally yields the pseudo residuals. With the Bernoulli loss in equation 3.15 follows

$$e^{[m-1]} = -\frac{\partial L_B(y, g(\mathbf{x}))}{\partial g(\mathbf{x})} \bigg|_{g(\mathbf{x})=g^{[m-1]}(\mathbf{x})} = y - \frac{1}{1 + \exp(-g^{[m-1]}(\mathbf{x}))} = y - f^{[m-1]}(\mathbf{x}) . \quad (3.16)$$

The conditional probability is optimized with each iteration m by building a new tree $b^{[m]}(\mathbf{x})$ that minimizes the previous error in equation 3.16 with respect to the true outcome.

One can iteratively updated an initialized sequence until convergence or until attaining M . Following Friedman (2001) to summarize the procedure, let $g^{[0]}(\mathbf{x}) = b^{[0]}(\mathbf{x})$ initialize the sequence with the mean of the target in the dataset $\mathcal{D}$, i.e. the proportion of the positive class, which yields $e^{[0]}$. Based on that each tree $b^{[m]}(\mathbf{x})$ at $m \in \{1, 2, \dots, M\}$ is fitted to the pseudo residuals $e^{[m-1]}$ that are obtained from the gradient of the objective function evaluated

at $g^{[m-1]}(\mathbf{x})$. Here, each tree aims on minimizing the pseudo residuals which implies an improvement in the fit and hence the predicted conditional probabilities as shown in equation 3.16. This yields the complete sequence $g^{[M]}(\mathbf{x})$ as final model

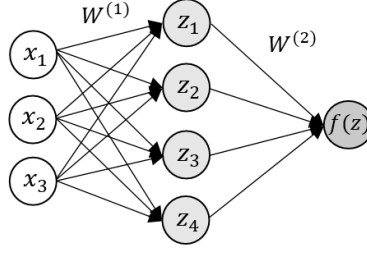
$$f_{GB}(\mathbf{x}) = \frac{1}{1 + \exp(-g^{[M]}(\mathbf{x}))} \quad \text{where} \quad g^{[M]}(\mathbf{x}) = b^{[0]}(\mathbf{x}) + b^{[1]}(\mathbf{x}) + \dots + b^{[M]}(\mathbf{x}) . \quad (3.17)$$

In fact, there is a need for complexity control since GB aggressively minimizes the loss (Hastie et al. (2009), p. 362). This can lead to overfitting, see Figure 2.2. Two possible hyperparameters become apparent from the sequence $g^{[M]}$ in equation 3.17. Those are the maximum number of iterations M in terms of boosted trees and the maximum depth for the trees $b^{[m]}(\mathbf{x})$. However, further adjustments for complexity can be made directly in $b^{[m]}(\mathbf{x})$ by imposing a penalty on the step length along the steepest descent (Friedman (2001), p. 1203). In order to implement that, a learning rate $\eta \in (0, 1]$ is introduced such that the fitted tree becomes $\forall m$ the scaled version $\eta \cdot b^{[m]}(\mathbf{x})$. The scaling applies to the tree outputs and thus shrinks the contribution of each tree to the whole ensemble. This slows down the learning process. However, setting and altering η can help with an improved convergence to a minimum. The maximum number of iterations, the maximum depth and η are hyperparameters that aim on changing the complexity. Friedman (2001) finds a dependency between complexity parameters and hence those have to be identified simultaneously.

The implementation of Chen & Guestrin (2016) used in chapter 5 makes several adjustments with regards to efficiency. In particular, the second order Taylor approximation of the objective function as introduced in Friedman et al. (2000) and accelerated split finding by approximate histograms are the most extensive modifications.

3.3 Feed-forward Neural Network

Further candidates for algorithms are Neural Networks. They can be described in detail by directed graphs where both the nodes and the directed edges as connectors have a specific purpose. Figure 3.4 represents the simplest neural network form. It feeds the input, without any acyclic connections, forward to the output. The structure is also called Multi-Layer Perceptron (MLP) (Bishop (1994), p. 1808). Figure 3.4 has one input, hidden and output layer whose sizes k , D and L are in general variably adjustable. The two sets of edges connecting the layers are the $(k \times D)$ weight matrix $W^{(1)}$ and, depending on the output layer size, the weight vector or matrix $W^{(2)}$ of dimension $(D \times L)$. Due to practical relevance in chapter 5, the hyperbolic tangent is considered as an activation function $a(\cdot)$ of the hidden layer. With $d \in \{1, 2, \dots, D\}$ and single weights w , the mapping from the input to the hidden

Figure 3.4: Exemplary Feed-forward Neural Network with $k = 3$, $D = 4$ and $L = 1$

layer follows by

$$z_d = a(v_d) = \frac{\exp(v_d) - \exp(-v_d)}{\exp(v_d) + \exp(-v_d)} \quad \text{where} \quad v_d = w_{0d}^{(1)} + \sum_{j=1}^k w_{jd}^{(1)} x_j. \quad (3.18)$$

The activation function of the hidden layer in equation 3.18 induces non-linearity and it is shifted by the bias term $w_0^{(1)}$. The output layer activation function is determined by the problem at hand. Predicting conditional probabilities over two classes implies the use of the logit-transformation as in equation 3.5 or equation 3.17, which is referred to as sigmoid function. With $L = 1$, this reduces to

$$f_{MLP}(z) = \frac{1}{1 + \exp(-(w_0^{(2)} + \sum_{d=1}^D w_d^{(2)} z_d))}. \quad (3.19)$$

Note that the set of weights $W = \{W^{(1)}, W^{(2)}\}$ represents parameters θ that have to be estimated in order to yield a model that optimizes the predictive performance. This implies the consideration of the BC from equation 2.3 as loss function to minimize the associated empirical risk $R_{emp}(W)$. Under the conditions described at the beginning of subsection 3.2.2, one can use gradient descent to train a neural network. The updates are a sequential reassignment by

$$W^{[m]} \leftarrow W^{[m-1]} - \eta \cdot \left. \frac{\partial R_{emp}(W)}{\partial W} \right|_{W=W^{[m-1]}} \quad (3.20)$$

where η is again the learning rate and $W^{[0]}$ are initialized by random draws from an uniform distribution (Glorot & Bengio (2010), p. 251). The procedure in equation 3.20 differs from Boosting, since the gradient updates trace back to the first weights via equation 3.19 and the activation function $a(\cdot)$. This is incorporated in the computation of the gradient by the

chain rule. For the influence of the each weight on the risk minimum follows generally

$$\frac{\partial R_{emp}(W)}{\partial W^{(2)}} = \frac{\partial R_{emp}(W)}{\partial f_{MLP}} \cdot \frac{\partial f_{MLP}}{\partial W^{(2)}} \quad \text{and} \quad \frac{\partial R_{emp}(W)}{\partial W^{(1)}} = \frac{\partial R_{emp}(W)}{\partial f_{MLP}} \cdot \frac{\partial f_{MLP}}{\partial \mathbf{z}} \cdot \frac{\partial \mathbf{z}}{\partial W^{(1)}} \quad (3.21)$$

which can be done column-wise for every weight vector in $W^{(1)}$ and $W^{(2)}$ with the gradient evaluated at the former weights $W^{[m-1]}$. The training of MLP via equation 3.21 called Back-Propagation.

For illustration, one can take BC from equation 2.3 as loss function and equation 3.19 as well as underlying equation 3.18 with the identity function as $a(\cdot)$ in vector notation. Following Bishop (1994) and taking the derivative of the loss function with respect to the weights, a theoretic steepest descent update step for one column in $W^{(1)}$ becomes

$$\begin{aligned} \frac{\partial L_{BC}(y, f_{MLP}(\mathbf{z}))}{\partial W_d^{(1)}} &= - \left(-\frac{y}{f_{MLP}(\mathbf{z})} + \frac{1-y}{1-f_{MLP}(\mathbf{z})} \right) \cdot f_{MLP}(\mathbf{z})(1-f_{MLP}(\mathbf{z})) \cdot \mathbf{x} \\ &= (y - f_{MLP}(\mathbf{z})) \mathbf{x} . \end{aligned} \quad (3.22)$$

The three derivatives in equation 3.22 follow from the chain rule in equation 3.21. They stem from BC together with equation 3.19, the sigmoid function in equation 3.19 and the initial linear combination of weights in equation 3.18. Each update optimizes the conditional probability prediction. That the gradient updates in equation 3.22 depend on $\mathbf{x}$ and its scale. A common scale of the explanatory variables becomes beneficial for smoother convergence.

Since the computation of the updates can be costly, the dataset $\mathcal{D}$ is partitioned into batches b such that an update with the complete dataset information becomes $\frac{\partial R_{emp}(W)}{\partial W} = \frac{1}{B} \sum_{b=1}^B \frac{\partial R_{emp}^b(W)}{\partial W}$. Updating the weights iteratively and considering each batch once, finishes one epoch. Drawing batches in each epoch randomly without replacement, results in a stochastic version of gradient descent as described in Bottou (2010). The amount of learned epochs can be determined using validation by early stopping. In line with equation 2.10 and section 5.4, the neural network is trained epochs until the outer loss on a validation set stops improving. Further, sticking to the default batch sizes of the implementations used in chapter 5 and the hyperbolic tangent as activation function $a(\cdot)$, the most relevant hyperparameters to optimize are the size of the hidden layer and the learning rate η .

4 Multi-label Approaches

Following the theoretical extension to the multi-label framework in section 2.3, this chapter introduces candidate multi-label approaches. With one benchmark approach that assumes label independence, the emphasis is on approaches with probabilistic output that are theoretically able to capture dependencies between the targets. Here, Tsoumakas & Katakis (2007) introduces two key concepts that are called Problem Transformation and Algorithm Adaptation. The former rearranges the problem to a binary or multi-class prediction framework and solves it by utilizing the probabilistic single target algorithms of chapter 3. The latter adapts suiting probabilistic base algorithms of chapter 3 to enable them to predict the joint multi-label probability distribution directly and capture dependencies during the learning process. Further, there exist several homogeneous as well as heterogeneous ensembles which form their own third approach. Those utilize either Bagging as introduced at the beginning of subsection 3.2.1 or Stacking established by Wolpert (1992) to improve the predictive performance. The three approaches are introduced in the following.

4.1 Problem Transformation

The aim of Problem Transformation is to rephrase the multi-label problem into a more established framework such that the base algorithms of chapter 3 are directly applicable. Figure 4.1 depicts approaches which state binary problems in parallel, transform the label combinations into classes or identify dependent binary problems sequentially.

Algorithm (a) in Figure 4.1 resembles the baseline approach. The label space is decomposed into L independent binary problems and a probabilistic base algorithm is trained for $y_l \forall l$ without considering any dependency between the labels (M.-L. Zhang & Zhou (2013), p. 13 f.). This algorithm is called Binary Relevance (BR). Equation 2.13 reduces to

$$h_{BR}(\mathbf{x}) = [h_1(\mathbf{x}) \ h_2(\mathbf{x}) \ \dots \ h_L(\mathbf{x})] \quad \text{where} \quad h_l(\mathbf{x}) = f_l(\mathbf{x}) = P(y_l = 1|\mathbf{x}) \quad (4.1)$$

by assuming conditional independence as described in equation 2.11. Note that $f_l(\mathbf{x})$ uses the same databases $\forall l$ in equation 4.1. An optimized and empirical risk minimal base algorithm has to be identified with cross-validation, which can be used to determine $P(y_l = 1|\mathbf{x}) \forall l$.

Algorithm (b) in Figure 4.1 is called Label Powerset (LP) and introduced in Tsoumakas & Katakis (2007) as transformation of the multi-label problem into a single multi-class problem.

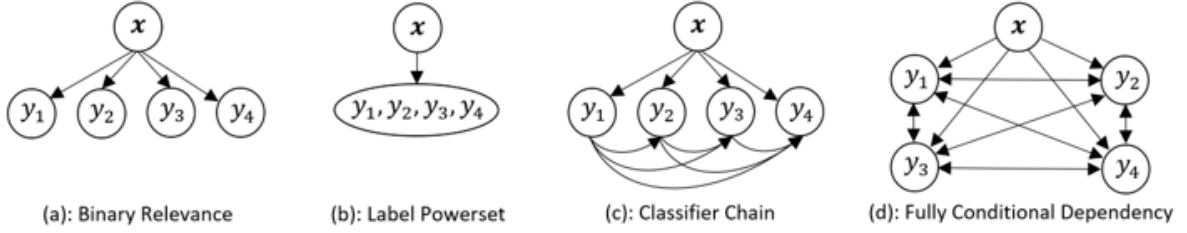


Figure 4.1: Schematic illustration of Problem Transformation approaches with four targets (Ceylan & Pekel (2017), p. 233)

Each of the 2^L possible label combinations that occurs in $\mathbf{Y}$ is relabeled and considered its own class (Read et al. (2014), p. 2 f.). As the number of classes grows exponentially in the number of targets, the available instances per class for algorithm training lowers rapidly. This becomes more severe as the number of imbalanced targets increase and hence this algorithm is only suited for low dimensional target spaces. The LP algorithm capture dependencies, since the label combinations are trained together in classes. With the use of a probabilistic multi-class algorithm, the underlying conditional joint distribution is determined. Note that the resulting multi-class problem is a generalization of a single label problem with two classes. Accordingly, the probabilistic algorithms in chapter 3 need to be extended from the binary to the multinomial framework. Due to relevance in chapter 5, this is shown at the example of LR, where the logit-transformation in equation 3.5 becomes

$$P(G = b|\mathbf{x}) = \frac{\exp(\beta_{0b} + \mathbf{x}\tilde{\beta}_b)}{\sum_{c=1}^C \exp(\beta_{0c} + \mathbf{x}\tilde{\beta}_c)} . \quad (4.2)$$

G is the multi-class target with classes $c, b \in \{1, 2, \dots, C\}$ (Friedman et al. (2010), p. 9). b is the respective class of interest with one class being excluded as reference. Still, it is uniquely identified by $\sum_{c=1}^C P(G = c|\mathbf{x}) = 1$. This leads to the $C - 1$ classes b can take on in equation 4.2. Extending the BC loss from equation 2.3 to the Cross-Entropy loss yields

$$\mathcal{L}_{CE}(G, f(\mathbf{x})) = - \sum_{c=1}^C G_c \log(P(G = c|\mathbf{x})) \quad (4.3)$$

where G_c is an indicator $\mathbf{1}_{\{G=c\}}$ that is one if c is the true class and zero else (Friedman et al. (2010), p. 10). Substituting the BC loss from section 3.1 by equation 4.3, one can adapt the penalized log-likelihood in equation 3.6 to the multi-class case. Equivalent to the computational steps in section 3.1, Friedman et al. (2010) optimize the resulting penalized weighted least-square by additionally considering each class. However, a more efficient incremental gradient solver is given in Defazio et al. (2014) and also used for chapter 5. After the model identification, all class probabilities for some instance $\mathbf{x}$ can be used to obtain the marginal probabilities. This is done by a reverse mapping of the predictions for the classes c

to the underlying label set. More generally the algorithm turns equation 2.13 to

$$h_{LP}(\mathbf{x}) = [h_1(\mathbf{x}) h_2(\mathbf{x}) \dots h_L(\mathbf{x})] \quad \text{where} \quad h_l(\mathbf{x}) = \sum_{c=1}^C \mathbf{1}_{\{(y_l=1) \in c\}} P(G = c | \mathbf{x}) . \quad (4.4)$$

The marginal probabilities $P(y_l = 1 | Dep(y_l), \mathbf{x})$ are $\forall l$ given by summing over all class probabilities containing the positive class label of the respective target in equation 4.4.

Algorithm (c) in Figure 4.1 is called Classifier Chain (CC). It is introduced in Read et al. (2009) and reformulated into a probabilistic framework by Dembczyński et al. (2010). The probabilistic version of CC is of primary interest here. It is derived from the theoretic decomposition of the conditional joint distribution $P(\mathbf{y} | \mathbf{x})$ by the product rule as in equation 2.12. A probabilistic base algorithm f predicts one marginal distribution over an initial target y_1 given only $\mathbf{x}$. This predictions are then used to augment the input matrix for the prediction of the marginal distribution of the next target y_2 and so forth. The input matrix is iteratively augmented by up to $L - 1$ predictions which implies the estimation of L probabilistic base algorithms (Dembczyński et al. (2010), p. 4). This results in

$$h_{CC}(\mathbf{x}) = [h_1(\mathbf{x}) \dots h_L(\mathbf{x})] = [P(y_1 = 1 | \mathbf{x}) \dots P(y_L = 1 | \mathbf{x}, y_1, \dots, y_{L-1})] . \quad (4.5)$$

Apart from the used base algorithm, the performance can depend on the order of the targets in the chain which can be determined by empirical risk minimization (Dembczyński et al. (2010), p. 4). This implies that every permutation has to be evaluated for equation 4.5. The complexity of this rises in the number of target variables with 2^L possible paths.

Algorithm (d) in Figure 4.1 is called Fully Conditional Dependency (FCD). It is inspired by Montanes et al. (2014) and van Buuren et al. (2006). Instead of focusing on classification like in the former, a probabilistic perspective is taken. The iterative approach of the latter is utilized together with a probabilistic base algorithm. Similar to the CC, the marginal distributions are predicted sequentially. However, all the respective other predictions are now conditioned on. This requires $\forall l$ an initialization $y_l^{[0]}$ which is done by the means of BR from equation 4.1. This is followed by $m \in \{1, 2, \dots, M\}$ iterations through

$$h_{FCD}(\mathbf{x}) = [h_1(\mathbf{x}) \dots h_L(\mathbf{x})] = [P(y_1^{[m+1]} = 1 | \mathbf{x}, y_{-1}^{[m]}) \dots P(y_L^{[m+1]} = 1 | \mathbf{x}, y_{-L}^{[m+1]})] \quad (4.6)$$

with $P(y_i^{[m+1]} = 1 | \mathbf{x}, y_1^{[m+1]}, \dots, y_{i-1}^{[m+1]}, y_{i+1}^{[m]}, \dots, y_L^{[m]})$ for any interim target. Note that y_{-1} incorporates all targets except y_1 . Senge et al. (2019) find that the use of probability transformed labels in the training set is beneficial. This is illustrated in Figure 4.2. Here, the situations during training and prediction match for the auxiliary labels and the data distributions remains equal reducing distortion. Accordingly, a conditional probability is used in $P(y_1^{[m+1]} = 1 | \mathbf{x}, h_{-1}(\mathbf{x})^{[m]})$ instead of the true labels. For all training data labels,

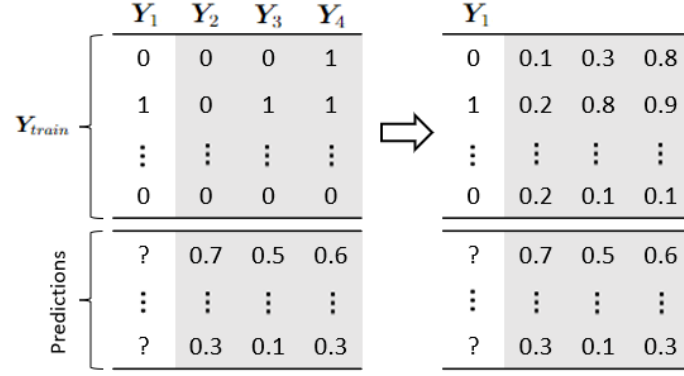


Figure 4.2: Transformation of the auxiliary training set labels

cross-validated predictions of the conditional probabilities are generated with BR from equation 4.1 once and then auxiliary used in equation 4.6. With that transformation, equation 4.6 iteratively updates the predictions by rotation of j through the identified chain order considering prior predictions for the test data targets as data augmentations. This is deterministic by neglecting parameter draws from their distributions at each iteration, which is the major difference to Gibbs-Sampling used in van Buuren et al. (2006). Beside the ordering, the number of iterations M are of relevance. One can run 4.6 several times until the empirical risk minimum is reached. However, for the analysis conducted in chapter 5 the iterations are heuristically set to 10.

The approaches introduced in Figure 4.1 reformulate the multi-label problem such that the probabilistic base algorithms in chapter 3 become applicable. If the probabilistic base algorithms are instead reformulated to solve the multi-label problem directly, one yields the Algorithm Adaptation approaches.

4.2 Algorithm Adaptation

Note that existing adaptations of algorithms are often domain specific and hence are not applicable in general settings (Read (2010), p. 58). In fact, the majority of developed approaches come from field of natural language processing or computer vision (Herrera et al. (2016), p. 81 ff.). Further, the uncommon probabilistic perspective lowers the availability of algorithms, hence only two simple adaptations to the probabilistic base algorithms are considered. Those are a multi-label versions of RF and the MLP.

The adaptation of RF, which is originally proposed by Clare & King (2001), takes place at the splitting criterion. Note that the used implementation is based on Dumont et al. (2009), which is similar up to a normalizing constant. The adapted version allows the evaluation of the candidate splitting points across all targets in $\mathbf{y}$ simultaneously. This is done by considering the average of the target impurity measures in the subsets $\mathcal{D}_1$ and $\mathcal{D}_2$ and further

at each resulting node that is generated by a split. Following equation 3.9, one is again trying to achieve the maximal impurity reduction but with respect to $\mathbf{y}$. Due to the higher dimensionality of $\mathbf{y}$, the impurity is now identified by

$$I_{ML}(\mathcal{D}_1) = -\frac{1}{L} \sum_{l=1}^L \pi_l \log(\pi_l) - (1 - \pi_l) \log(1 - \pi_l) \quad \text{with} \quad \pi_l \in [0, 1] . \quad (4.7)$$

In equation 4.7, π_l is the proportion of the positive class of target y_l in the respective node. Other aspects equal to subsection 3.2.1, a evaluation of the splits for $\mathbf{x}$ result in leaf q and

$$h_l(\mathbf{x}) = \frac{1}{M} \sum_{m=1}^M f_l^{[m]}(\mathbf{x}) \quad \text{with} \quad f_l^{[m]}(\mathbf{x}) = \frac{1}{n_q^{[m]}} \sum_{(\mathbf{x}_i, y_{il}) \in \mathcal{D}_q^{[m]}} \mathbb{1}_{\{y_{il}=1\}} . \quad (4.8)$$

q is one of Q leafs with cardinality n_q , in which the training data instances $\mathcal{D}_q^{[m]}$ are allocated. Each instance $\mathbf{x}$ terminates in one leaf q for each tree m , in an ensemble of size M . This yields conditional probabilities as proportions of the positive class of y_l in $\mathcal{D}_q$. With an ensemble aggregation by averaging $\forall l$ in equation 4.8, a Multi-label Random Forest (MLRF) is defined by the output vector $h_{MLRF}(\mathbf{x}) = [h_1(\mathbf{x}) \dots h_L(\mathbf{x})]$. The same hyperparameters as in subsection 3.2.1 have to be optimized.

The main adjustment of MLP to the multi-label framework, is the consideration of L nodes in the output layer of Figure 3.4. In line with equation 3.19, each of the L output node is activated by its own sigmoid function. This implies an adjustment to the loss function. As Nam et al. (2014) show, the BC loss from equation 2.3 can be extended and combined to the MBC loss of equation 2.14. One can consider this as Neural Network equivalent to BR since the weights $W_l^{(2)}$ can be trained by independent updates $\forall l$ (He & Xia (2018), p. 1). However, $W^{(1)}$ is trained with respect to all output nodes via the common hidden layer. Sticking to Figure 3.4 and the underlying notation, Back-Propagation of the first weight set becomes

$$\frac{\partial R_{emp}(W)}{\partial W^{(1)}} = \sum_{l=1}^L \frac{\partial R_{emp}(W)}{\partial f_l} \cdot \frac{\partial f_l}{\partial \mathbf{z}_l} \cdot \frac{\partial \mathbf{z}_l}{\partial W_l^{(1)}} \quad (4.9)$$

by the chain rule for partial derivatives (Bishop (1994), p. 1812). Due to the fact that all output nodes depend on the hidden layer, the updates to $W^{(1)}$ in equation 4.9 are based on all targets L , which captures signals coming from all targets during training. Existing dependencies between the targets are implicitly incorporated. This effect increases by adding a further intermediate hidden layer to the neural network structure in Figure 3.4. Here, each node incorporates the hyperbolic tangent as activation function and a matrix of weights connecting the two hidden layers $Z^{(1)}$ and $Z^{(2)}$ is added. Having $W = \{W^{(1)}, W^{(2)}, W^{(3)}\}$,

the added weights are trained by the chain rule for partial derivatives and Back-Propagation. This results in a Deep Neural Network (DNN). Further, the additional layer allows the neural network to adapt to higher dimensional interactions and form arbitrary complex decision boundaries (Lippmann (1987), p. 1812). As adverse side effect, deep neural network structures are more prone to overfitting. This can be dealt with by the consideration of dropout, which decouples the hidden nodes by randomly deactivating a proportion of edges in Figure 3.4 that correspond to the weights (Nam et al. (2014), p. 8). Deactivation implies that the respective weights are temporarily disregarded in the updates which simultaneously lengthens that training process. Hinton et al. (2012) show that this functions as regularization and hence avoids overfitting.

$$h_{DNN}(\mathbf{x}) = [h_1(\mathbf{x}) \dots h_L(\mathbf{x})] \quad \text{where} \quad h_l(\mathbf{x}) = \frac{1}{1 + \exp(-(w_{0l}^{(3)} + \sum_{d=1}^D w_{dl}^{(3)} z_d^{(2)}))} \quad (4.10)$$

defines the used deep multi-label version of the MLP. For equation 4.10, the number of nodes in the added hidden layer as well as the fraction of edges to be deactivated by dropout have to be identified as additional hyperparameters compared to section 3.3. This is done via early stopping by attaining the empirical risk minimum of the validation loss.

4.3 Approach Ensembles

The last approach consists of ensembles of Problem Transformation approaches together with varying base algorithms from chapter 3. In fact, the majority of ensembles are proposed for specific Problem Transformation algorithms in the literature and accordingly introduced. Bagging as well as Stacking are utilized in homogeneous and heterogeneous ensembles.

Wolpert (1992) introduces Stacking as a heterogeneous ensemble. For the problem at hand, this implies that a single Problem Transformation approach is utilized together with several varying probabilistic base algorithms by combining the respective target predictions to a meta dataset in the first stage. On this meta dataset, a Problem Transformation approach with a probabilistic meta algorithm is retrained with respect to the target variables as second stage. Let the Problem Transformation approach be BR from equation 4.1 at all stages and the exemplary probabilistic base algorithms be LR from section 3.1 as well as GB from subsection 3.2.2. In that case the Heterogeneous Stacking Ensemble (HSE) becomes

$$h_{HSE}(\mathbf{x}) = \left[h_1([h_{BR-LR}(\mathbf{x}) \ h_{BR-GB}(\mathbf{x})]) \dots h_L([h_{BR-LR}(\mathbf{x}) \ h_{BR-GB}(\mathbf{x})]) \right] \quad (4.11)$$

with $h_{BR-LR}(\mathbf{x}) = [f_1^{LR}(\mathbf{x}) \dots f_L^{LR}(\mathbf{x})]$ and $h_{BR-GB}(\mathbf{x}) = [f_1^{GB}(\mathbf{x}) \dots f_L^{GB}(\mathbf{x})]$. h_l is $\forall l$ the probabilistic meta algorithm where Ting & Witten (1999) identify LR from section 3.1 to be suiting. Note that every true label instance of the training data is, identical to Figure 4.2,

transformed into conditional probabilities in the meta dataset using cross-validation. This keeps data distributions for training and prediction equal. With that, equation 4.11 can learn the strengths and weaknesses of the underlying probabilistic base algorithms. Further, dependencies between the targets are incorporated by training the probabilistic meta algorithm on the L conditional probability distributions of each probabilistic base algorithm. With respect to the analysis conducted in chapter 5, BR is utilized at all stages. At the first stage, all probabilistic base algorithms from chapter 3 are considered. The target predictions are combined to a meta dataset where the true training labels are transformed to conditional probabilities. Based on this meta dataset, LR is used as meta prediction algorithm.

The idea for a simple homogeneous Stacking ensemble is introduced in Godbole & Sarawagi (2004). Stacking is used to augment the input space. The approach is used initially conditioned on $\mathbf{x}$ alone. Following The probabilistic predictions are then used to augment the dataset. Following Figure 4.2, the training set labels are again transformed to conditional probabilities using cross-validation. Without being selective about the information added, the same approach is then used to append new predictions to augment the dataset further and so forth (Montanes et al. (2014), p. 1496). This can be continued until the predictions stop improving. Let the utilized approach be BR, then follows Binary Relevance Stacking (BRS) with an arbitrary probabilistic base algorithm as

$$h_{BRS}^{[2]}(\mathbf{x}) = \left[h_{BR}^{[2]}(\mathbf{x}, h_{BR}(\mathbf{x}), h_{BR}^{[1]}(\mathbf{x}, h_{BR}(\mathbf{x}))) \right] \quad (4.12)$$

where the number of stacks is $M = 2$ for illustrative reasons. For the sake of the analysis in chapter 5, the number of stacks M of equation 4.12 is heuristically set to 5.

Besides the introduction of CC, Read et al. (2009) suggest a homogeneous Bagging ensemble. This is based on random sub-sampling, i.e. draws from $\mathcal{D}$ without replacement, instead of bootstrapping and increased variability by random ordering in each chain. Aggregation is conducted by averaging the probability predictions across all chains which results in an Ensemble of Classifier Chains (ECC). Further and similar, Read et al. (2009) define an Ensemble of Binary Relevance (EBR), where the variability stems only from the random sub-sampling. Those two ensembles are given by

$$h_{ECC}(\mathbf{x}) = \frac{1}{M} \sum_{m=1}^M h_{CC}^{[m]}(\mathbf{x}) \quad \text{and} \quad h_{EBR}(\mathbf{x}) = \frac{1}{M} \sum_{m=1}^M h_{BR}^{[m]}(\mathbf{x}) \quad (4.13)$$

where the variability of the CC and BR across all m is utilized by Bagging. Following Read et al. (2009), the randomly sub-sampled training data proportions at m are set to 67% and an ensemble size with $M = 10$ is used for both ensemble approaches in equation 4.13.

An Heterogeneous Bagging Ensemble (HBE) is introduced in Tahir et al. (2012). The required variability stems from the use of different approaches and especially from alternation

of the probabilistic base algorithms. Those solve the prediction problem in a diverse manners. By the combination of the predictions one can take, similar to equation 4.11, advantage of the specific algorithm strengths. Here, averaging the predicted probabilities yields the best improvements in discrimination (Tahir et al. (2012), p. 515 ff.). The HBE is given by

$$h_{HBE}(\mathbf{x}) = \frac{h^{[1]}(\mathbf{x}) + h^{[2]}(\mathbf{x}) + \dots + h^{[M]}(\mathbf{x})}{M} \quad (4.14)$$

where $h^{[m]}(\mathbf{x})$ is $\forall m$, either by the underlying probabilistic base algorithm or the multi-label approach used, unique in the ensemble. The HBE in equation 4.14 does not make use of dependencies between the targets, but instead captures differences in the predictions of the underlying algorithms and approaches advantageously. Apart from the selection of $h^{[m]}(\mathbf{x})$ $\forall m$, no further hyperparameter has to be considered here.

5 Methodology

Based on the theoretic foundations of chapter 2, the probabilistic base algorithms in chapter 3 and the multi-label approaches of chapter 4, this chapter explains the decisions and computational steps taken to address the raised problems and the goals of this thesis as outlined in chapter 1. Firstly, relevant peculiarities of the survey and its methodology for the data collection are introduced. This is followed by a section that illustrates the survey data preprocessing steps. Those are target and explanatory variable selection, imputation of missing data, dataset partitioning by splits as well as encoding of categories and data scaling. After data preprocessing, algorithms and approaches are applicable to the data. A section on performance metrics of interest, i.e. with respect to the discriminative capabilities of the predicted probability distributions, follows. Those allow for the assessment of the predictive performance of the multi-label approach and make them comparable. Finally, the identification of optimized specifications of probabilistic base algorithms and selected multi-label approaches as well as the experimental setups for the evaluation are discussed. Optimization implies choosing hyperparameter combinations such that the expected generalization error of the MBC loss in equation 2.16 is minimized by validation. The evaluation is conducted, with varying sophistication, for the data preprocessing steps, the optimization of the probabilistic base algorithms and the multi-label approaches. The emphasis is on the latter.

Note that the majority of the computational part is conducted with the programming language `Python`. The *Anaconda Software Distribution* (2020) is used which entails several software pre-installations. This includes `Jupyter Notebook` by Kluyver et al. (2016), which is used for coding and documentation. General data management is done with `pandas` by McKinney (2010) and numerical operations rely on `NumPy` by Harris et al. (2020). The computations for data preprocessing, probabilistic base algorithms, multi-label approaches and performance metrics are largely based on `scikit-learn` by Pedregosa et al. (2011). According visualizations are generated with `matplotlib` by Hunter (2007). Beside own implementations, some algorithms and approaches are generated using `TensorFlow` by Abadi et al. (2015), `scikit-multilearn` by Szymanski & Kajdanowicz (2019) and `scikit-multiflow` by Montiel et al. (2018). Note that `R` is only used for the imputation of missing data via the `RStudio` environment by RStudio Team (2020).

5.1 The German Health Update

The data stem from the German Health Update (GEDA). It is a cross-sectional health survey that is conducted by the Robert Koch Institute by interviewing a representative sub-sample of the German population. The institute researches in commission of the German Federal Ministry of Health. This chapter introduces the GEDA and summarizes the survey methodology as well as the data generating process. The summary is based on the more detailed descriptions given in Saß et al. (2017) and Lange et al. (2017).

The primary intend of the survey is the monitoring of the health condition within the German population. This is generally done via the collection and use of representative survey data that allow inference on the population level. This is, for example, done for the identification of a disease prevalence. The required data is generated via questionnaires and collected in irregular waves. The available dataset stem from the fourth wave collected sequentially from November 2014 to July 2015. The total sample was divided into two tranches, autumn to winter 2014 as well as spring to summer 2015, to prevent a seasonal bias. The respective questioning took place in a Mixed-Mode-Design. This implies that all participants where offered an online-based questionnaire in the first place. If participants did not submit the questionnaire within four weeks, they where allowed to choose from either a paper- or online-based questionnaire by providing a paper-based version in the reminding letter. A second reminder of the same kind was send after additional four weeks.

The sampling of candidates follows a two-stage stratified cluster sampling. In the first stage, 301 out of 11,339 communities were sampled. The sampling was stratified by 402 administrative districts as well as further factors like local and regional population size and employment density. Selected communities with fewer than 1000 inhabitants were combined with similar and small neighboring communities to form a single sample point. Major cities with large populations have been represented by multiple sample points. All identified sample points are illustrated in Figure 5.1. In the second stage, individuals in the selected locations are drawn from local population registers by stratified random sampling with respect to age. In order to conduct analysis for each of the 16 federal state levels and with sufficiently high precision, at least 67 participants were required per community as well as a minimum of 800, roughly gender equal, selected participants per federal state. Here, GEDA restricts the data to adults of age 18 or older with permanent residence in Germany. In total, the available data consists of 24,016 instances. Lange et al. (2017) find that the sample distributions already show a high level of representativeness. Nevertheless, a design and adjustment weights are provided that enables a further increase in representativeness. The design weight takes the selection probabilities of sample points as well as the selection probabilities of the subjects within the sample points into account. The adjustment weight aims on adjusting the willingness to participate which differs by population groups. The groups are identified

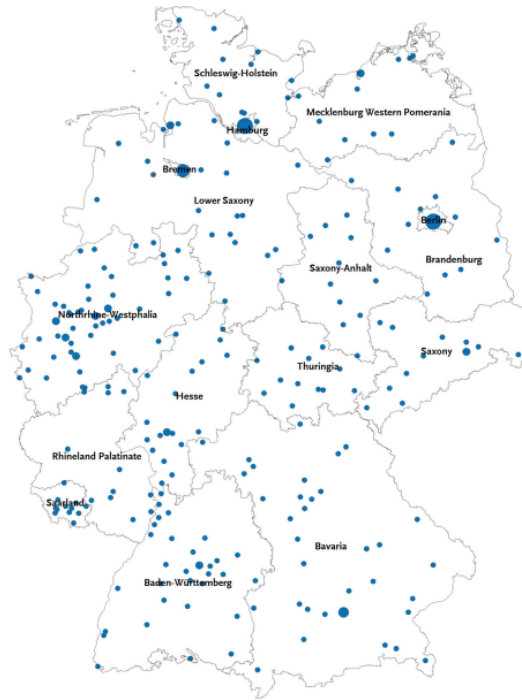


Figure 5.1: GEDA sample points (Lange et al. (2017), p. 3)

via age, gender, level of education as well as several geographic aspects. Combined, the two weights yields the GEDA weight that is provided together with the data. When a weighting procedure is applied, the sample distribution gets closer to the true distribution of the German population. This is invaluable for analyses at the national level.

The GEDA questionnaire consists of two components. These are domestically specific questions and augmentations for the European Health Interview Survey. The latter aims on provision of standardized and comparable health data, mainly for the states of the European Union. It serves as data basis for European health policy, health reporting and for the determination of European health indicators. Both components are considered equivalently here. The structure of the questionnaire can be divided in chapters with each chapter capturing several topics. Each of them consists of questions that represent variables of potential interest. An overview for chapters and topics is given by Figure 5.2. The chapter Technical Data is neglected, since it does not rely on questions. It contains the GEDA weights, sampling points as well as data that is directly determined by the sampling like date, federal state and regions. In total, GEDA consists of realizations from 269 variables with 15 apportioned to Technical Data. The 254 variables stemming from the questionnaire are allocated to the chapters in Figure 5.2. Here, 70 belong to General Health, 18 to Mental Health, 42 to Prevention, 53 to Health Behavior, 6 to Living Conditions and 65 to Personal Data. The former, General Health, contains 26 binary variables that belong to the topic Diseases. Thereof 22 indicate whether an individual had the respective disease within the

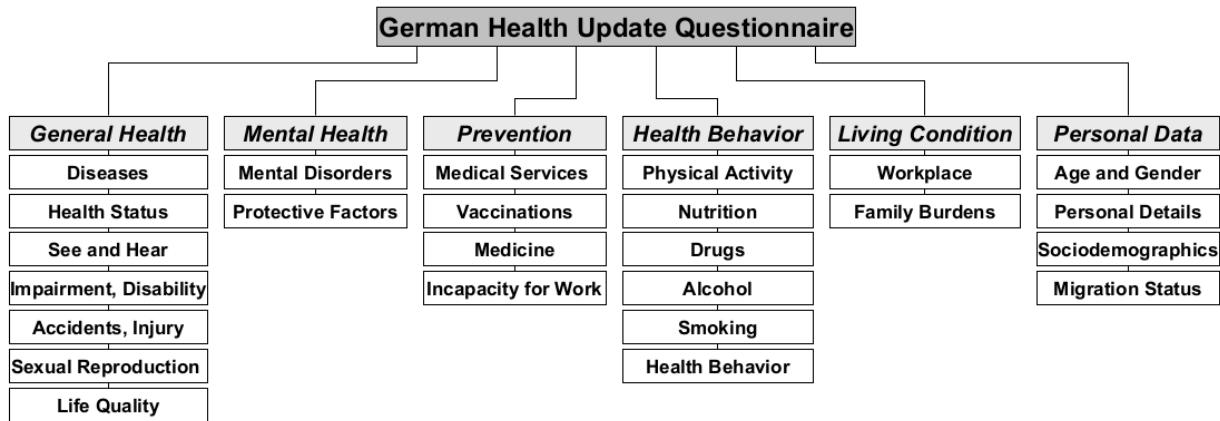


Figure 5.2: Chapters of the GEDA questionnaire with included topics

last 12 months or not. Those indicators are chosen as target candidates. The remaining topic variables and suiting variables from the Technical Data are considered as potential explanatory variables. The entire survey dataset follows from 24,016 survey participants with a maximum of 269 variable realizations per respondent due to possible missing values. The survey data preprocessing steps are outlined in the following.

5.2 Survey Data Preprocessing

The provided GEDA dataset does not allow for a direct application of the multi-label approaches and the underlying probabilistic base algorithms. This is especially caused by a substantial amount of missing data from item non-responses and the fact that categorical data, which the majority of the survey data consists of, are not processable by the majority of algorithms in chapter 3 and approaches of chapter 4. Beyond obligatory data preprocessing, further steps can be considered. The selected ones are illustrated in the data flowchart in Figure 5.3. This includes an initial selection of interdependent disease variables as targets and, based on this, the selection of the explanatory variables. It is followed by the multivariate imputation of item non-responses in the dataset. Afterwards, the general principle of data splits for the assessment of predictive performance is further discussed based on section 2.2. With the training and test set, the realizations of categorical variables are encoded into a numeric representation and the data is transformed to a common scale. The main objective of data preprocessing is to make algorithms and approaches applicable. Based on experimental setups, a predictive base algorithms is chosen to underlie the multi-label approaches and the approaches are assessed by their predictive performance. However, optimizing each preprocessing step can also have a general beneficial influence on the predictive performance. A comprehensive assessment is out of the scope of this thesis, still the majority of the chosen preprocessing methods is evaluated with respect to very simple approaches as benchmark

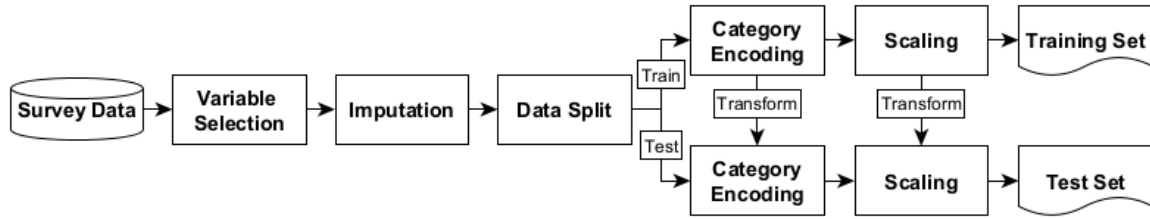


Figure 5.3: Preprocessing steps of the survey dataset

in section 6.1. Unlike the data preprocessing steps, data splits are a source of uncertainty for the evaluation of predictive performance. This is incorporated in the comparison of the multi-label approaches which is of primary interest and further discussed in section 5.4.

5.2.1 Target Variable Selection

The target variables are identified first. For the more general experimental setup, five interdependent diseases are selected by reasoning and empirical evaluation. Only variables from the topic Diseases are considered. They mainly represent diseases that are hard to identify without medical diagnosis where predictions are especially valuable. Note that mental disorders are not included since the related questions in the survey aim on a clear reconstruction of the diagnosis. The same is true for obesity where the disease definition of the World Health Organisation is based on the available Body-Mass-Index as combination of weight and height. In the topic Diseases, 20 variables are 12 month disease indicators. Here, self-reports of diseases are replaced with indicators based on medical diagnostic if available. Most of the 20 diseases have only a single-digit percentage of relative frequency of occurrence in the survey data. Relative frequency is indicated by the proportion with which a disease occurs in the dataset, i.e. the number of positive classes divided by the total number of available disease outcomes. Potential target candidates are required to display a relative frequency of at least 2.5%. This bound is chosen for several reasons. It is practical, since a comprehensive discussion of label imbalance is not part of the following sections. Accordingly, a relative frequency lower bound ensures that enough positive class example are available for algorithm training after splitting and during the resampling of the data. Moreover, a subset of diseases with higher relative frequency show a more harmonious interdependence across all diseases, while binary dependencies dominate the selection process in equation 5.2 with the consideration of lower relative frequency diseases.

In order to identify a combination of five interdependent target variables, the 12 disease variables with a relative frequency of at least 2.5% are evaluated according to their marginal dependency. In line with conditional dependence depicted in equation 2.12, Dembczyński et

al. (2012) state that the closely related marginal dependence is given by

$$P(Y_1, \dots, Y_L) = \int_{\mathcal{X}} P(Y_1, \dots, Y_L | X) d\mu(X) \quad \text{with} \quad P(Y_1, \dots, Y_L) \neq \prod_{l=1}^L P(Y_l) . \quad (5.1)$$

The former of equation 5.1 shows that the marginal dependency can be interpreted as expected dependency for the conditional dependency given all possible instances. However, one does not strictly imply the other (Dembczyński et al. (2012), p. 9). In order to test for the dependencies between the diseases based on the data, the chi-squared test by Pearson (1900) is used pairwise. This is described as multi-label variable selection technique in Spolaôr & Tsoumakas (2013), where the score of the test statistic is of interest. Note that all diseases variables only show binary realizations. The countable combinations in realizations for two disease variables reduce to true positives tp , false positives fp , false negatives fn and true negatives tn . Table 5.1 displays the underlying contingency table. With $e, l \in \{1, 2, \dots, 12\}$

$Y_e(\text{Disease})$	$Y_l(\text{Disease})$		Total
	Yes	No	
Yes	tp	fp	$(tp + fp)$
No	fn	tn	$(fn + tn)$
Total	$(tp + fn)$	$(fp + tn)$	$(tp + fn + fp + tn)$

Table 5.1: Contingency table for binary data

and $l \neq e$, tp counts when the positive classes of the target variables under evaluation co-occur in the realizations of the dataset, i.e. $y_l = 1$ and $y_e = 1$. fp counts cases, in which only the former variable shows a positive realization. tn and fn follows with the same logic, but with respect to the negative class. According to Spolaôr & Tsoumakas (2013), the chi-squared test statistic for binary variables, which is called the Chi-squared Score (CS), becomes

$$CS_{e,l}(tp, fn, fp, tn) = \frac{(tp - (tp + fp) \cdot p_{pos})^2}{(tp + fp) \cdot p_{pos}} + \frac{(fn - (tn + fn) \cdot p_{pos})^2}{(tn + fn) \cdot p_{pos}} + \frac{(fp - (tp + fp) \cdot p_{neg})^2}{(tp + fp) \cdot p_{neg}} + \frac{(tn - (tn + fn) \cdot p_{neg})^2}{(tn + fn) \cdot p_{neg}} \quad (5.2)$$

where $p_{pos} = \frac{(tp+fn)}{(tp+fp+fn+tn)}$ and $p_{neg} = \frac{(fp+tn)}{(tp+fp+fn+tn)}$ are marginal relative frequencies. Under H_0 the two diseases variables are stochastically independent given the disease realizations from the dataset. According to probability theory, two events are stochastically independent if the probability that they will occur together is equal to the product of the individual probabilities. This should be reflected empirically by the marginal relative frequencies and the occurrence counts of the possible realizations combinations. The numerators in equation 5.2 compares the actual amount of the outcome occurrences for tp , fp , fn and tn with the

expected under independence, i.e. according to the marginal relative frequencies. Dependency is indicated by deviations in the numerators of equation 5.2. An increase in the CS indicates stronger relations which is evidence for dependency (Spolaôr & Tsoumakas (2013), p. 3). Calculating the bi-variate CS scores for Y_l with respect to each disease variable Y_e with $e \neq l$, and taking the score average yields $\forall l$

$$\overline{CS}_l = \frac{1}{12-1} \sum_{e \neq l} CS_{l,e}(tp, fn, fp, tn) . \quad (5.3)$$

The five disease variables with the highest average CS are identified as multi-label targets with sufficient interdependence. An overview of the identified targets is given in Table 5.2. Here,

GEDA Name	Disease	Rel. Freq.	Non-Resp.	Avg. CS
KHhyp12pB	hypertension	30.80 %	0.20 %	616.02
KHhi12	heart failure	4.39 %	5.78 %	598.67
KHkhk12	coronary artery disease	3.78 %	5.59 %	562.45
KHdge12	arthrosis	16.23 %	5.26 %	554.60
KHlip12A	hyperlipidemia	19.89 %	0.67 %	354.34

Table 5.2: Overview of target variable realizations

the variable are named according to the GEDA dataset. The evaluation from equation 5.2 together with equation 5.3 identified hypertension, heart failure, coronary disease, arthrosis and hyperlipidemia as independent disease targets. The respective average CS are reported in Table 5.2. Moreover, the table displays the relative frequency of the positive class in the dataset as well as the proportion of item non-responses per disease. The other variables that belong to the topic Diseases in Figure 5.2, are dropped from the dataset and not used as explanatory variables or any other way in this analysis.

5.2.2 Explanatory Variable Selection

The selection of the explanatory variables is divided in a qualitative and quantitative assessment. The former rules out GEDA variables by reasoning.

Some variables are caused by the diseases. This implies the inclusion of an explanatory variable that captures information that is not available without knowledge of the diseases. Without further statistical test, those are identified by reasoning and screening of the available variable list for obvious candidates. Accordingly, the two variables **BBmyo12** and **BBsa12** are dropped from the dataset. Those stem from the topic Impairment and Disability and indicate chronic impairment after heart attack and stroke that are both included as variables in the survey topic Diseases. These variables require the respective disease to appear beforehand. Several other variables like **IAhypus**, **IAcholus** and **IAdiabus** are closely related to diagnostic.

They stem from the topic Medical Services and determine the time frame of the last blood pressure, blood lipid and blood glucose measurement. This medical consultation arguably increases with knowledge of the diseases as well. However, with primary interest in predictive performance of the multi-label approaches, only obvious candidates are removed.

Further, two equal gender variables are in the topic Personal Details. One is removed, since both carry exactly the same information. In line with that, several variables only vary in the levels of granularity that is addressed. Those are 10 variables concerning regions belonging to the Technical Data as well as 12 age group variables in the topic Personal Details. Here, the highest granularity variable, i.e. federal state and age by 5-year bins, can closely reconstruct the other variables. In the case of age groups, binning the highest granularity group can exactly match all other groups. Eventually, the binning of the highest granularity group can provide information gains, which can be seen as a simple form of Feature Engineering. However, only the variable with the highest granularity is retained, since the amount of new information gained by binning is considered small compared to the increase in dimensionality. In addition, the variables of the technical data that cannot be used directly for prediction are excluded. Those are the time stamps for month and year of the survey, the sample point stamp and the GEDA weights (`untm`, `untj`, `ppoint`, `wQS`, `wQSpop`).

An inclusion of all remaining explanatory variables is inadvisable due to computational cost and with regards to performance reduction by overfitting. Identifying sufficiently associated explanatory variables is a more suiting. A quantitative variable selection is conducted for the remaining 218 candidates. Similar to equation 5.2 and equation 5.3, this follows the chi-squared test by Pearson (1900). It is suiting, since the vast majority of the survey variables are categorical, i.e. of either nominal or ordinal scale. Even the few numeric variables show a low variability in their realization such that enough counts per realization are, without binning, available for the applicability of the chi-squared test. The test statistics for the realizations of each explanatory variable are calculated with respect to the realizations of all five target variables. Following Spolaôr & Tsoumakas (2013), the scores are averaged over the targets and variables with the highest average scores are selected. This resembles a frequently used variable selection technique for the multi-label framework (Kashef et al. (2018), p. 11). All explanatory variables are considered to be categorical, hence more than two category levels are possible. Accordingly, the test statistic in equation 5.2 is altered. Let the explanatory variable X_j have C category levels, then $CS_{l,j}$ is calculated with respect to target variable l and all categories $c = \{1, 2, \dots, C\}$. The rows in Table 5.1 are substituted and extended with the categories of X_j . Calculating the CS $\forall l = \{1, \dots, 5\}$ and averaging

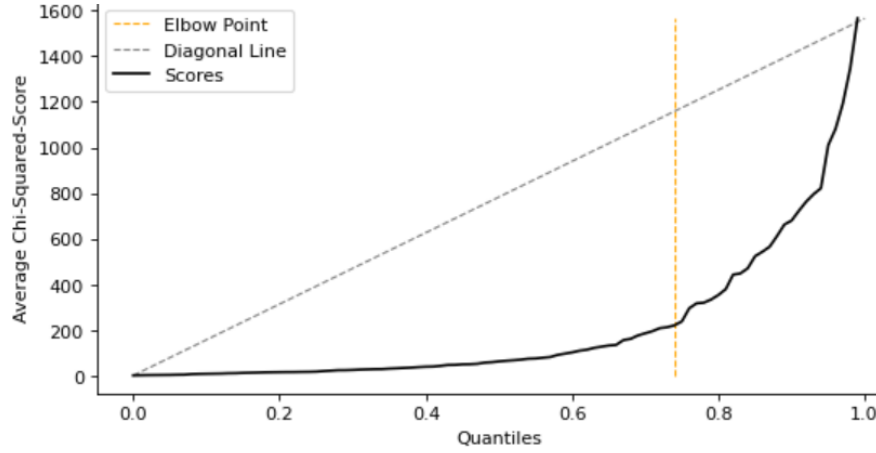


Figure 5.4: Heuristic explanatory variable selection by quantiles of the CS

the results yields $\forall j$

$$\overline{CS}_j = \frac{1}{5} \sum_{l=1}^5 CS_{j,l} \quad \text{where} \quad CS_{j,l} = \sum_{c=1}^C \sum_{b \in \{0,1\}} \frac{(n_{c,b} - n_{c,b}^*)^2}{n_{c,b}^*}. \quad (5.4)$$

This is a more general version of equation 5.3. $n_{c,b}$ counts the actual instances that show the combination of class b and category c . It is compared to $n_{c,b}^* = \frac{n_b \cdot n_c}{n}$ in the numerator of equation 5.4. This are the expected instances in the case of independence with n_b and n_c as marginal frequencies. For the selection of explanatory variables, a heuristic is used to identify a threshold value for the average CS in equation 5.4. It is inspired by the scree plot of Cattell (1966), that is based on the heuristic elbow method. The explanatory variables are identified by the trade-off in diminishing returns in dependency according to CS values and increasing computational cost and complexity when adding another explanatory variable. According to Satopaa et al. (2011), the trade-off quantile score is determined by the largest perpendicular distance between the scores and the diagonal line in Figure 5.4. Only explanatory variables with average CS exceeding this trade-off quantile score are considered and selected. This means that the average chi-squared value from equation 5.4 has to be higher than 220.39, the 74% quantile. The selected variables are given and described in Table 5.3. The 56 selected variables are ordered by their topic in the survey. Only five explanatory variables are numeric, ten are binary and the remaining are categorical. The reported average CS indicate which variables have the highest relevance with respect to the target diseases.

Note that the realizations of the explanatory variables as well as the realizations of the target variables in Table 5.2 have item non-responses. The proportion of the non-responses with respect to the total number of GEDA instances is displayed per variable in Figure 5.5. Here, two reasons for a missing value are possible according to the survey. Either an individual did not respond or the question did not apply due to the survey structure. The latter is, for

GEDA Name	Description	Scale	Topic	Avg. CS
age5B	Age in 5-year buckets	nominal	Age & Gender	2381.29
GZmehm1	General health condition	nominal	Health Status	1359.12
GZmehm3C	Chronic diseases indicator	binary	Health Status	1009.87
GZmehm2D	Restriction due to illness (at least 6 months)	nominal	Health Status	1139.44
SHseh1	Visual aid indicator	nominal	See & Hear	317.14
SHhoer1	Hearing aid indicator	nominal	See & Hear	320.02
SHhoer5	Hearing difficulties (quiet room)	nominal	See & Hear	343.01
SHhoer6	Hearing difficulties (noisy room)	nominal	See & Hear	353.34
BBdors112	Lower back complaints (last 12 months)	binary	Impairment,Disability	449.01
BBblas12B	Urinary incontinence (last 12 months)	binary	Impairment,Disability	360.53
BBgehen	Difficulties walking 500m unaided	nominal	Impairment,Disability	1010.06
BBtreppe	Difficulty with stairs (12 steps)	nominal	Impairment,Disability	1178.55
BBbehB	Officially acknowledged disability	binary	Impairment,Disability	798.25
LQsf367C	Severe pain (last 4 weeks)	nominal	Life Quality	684.66
LQsf368C	Disability due to pain (last 4 weeks)	nominal	Life Quality	795.30
IAhyypus	Last blood pressure measurement	nominal	Medical Services	459.99
IAcholus	Last blood lipid determination	nominal	Medical Services	1042.69
IAdiabus	Last blood glucose measurement	nominal	Medical Services	811.83
IAkosten3	Treatment not affordable (last 12 months)	nominal	Medical Services	315.77
IAarzt1B	Visits general practitioner	nominal	Medical Services	333.82
IAarzt1B4w	Number of visits (last 4 weeks)	ordinal	Medical Services	503.73
IAarzt1B12_k	Number of visits (last 12 months)	ordinal	Medical Services	604.82
IAkhs	Stationary in hospital (last 12 months)	binary	Medical Services	391.90
IAkhs_k	Nights stationary (last 12 months)	ordinal	Medical Services	542.15
IAkfutyp2B_lz	Last stool test	nominal	Medical Services	739.59
IAkfutyp4B_lz	Last colonoscopy	nominal	Medical Services	728.06
IAkfutyp7B_lz	Last mammogram	nominal	Medical Services	542.70
IPinfl4	Last flu shot	nominal	Vaccinations	656.30
IPinfl1415B	Flu vaccination winter 2014/2015	binary	Vaccinations	671.59
IPinfl1314B	Flu vaccination winter 2013/2014	binary	Vaccinations	638.27
IPinfl6B	Frequency of flu vaccination	nominal	Vaccinations	790.42
AMarztB	Prescribed Medication (within last 2 weeks)	binary	Medicine	1483.53
AUarbzD_k	Weeks ill & unable to work	nominal	Incapacity for Work	287.88
KAehispaq1	Work activities (exertion)	nominal	Physical Activity	713.09
RCex_ak	Age bucket when quitting smoking	nominal	Smoking	221.50
GVasku3	Good at solving exhausting tasks	nominal	Health Behavior	317.98
PAfam1C	Marital status	nominal	Personal Details	768.52
PAhh_k	Number of persons in household (HH)	ordinal	Personal Details	476.98
PAhhnbpersk_16_24	Number of 16 to 24 year olds in HH	ordinal	Personal Details	276.27
PAhhnbpersk_25_64	Number of 16 to 24 year olds in HH	ordinal	Personal Details	903.80
PAhhnbpersk_65+	Number of 65+ year olds in HH	ordinal	Personal Details	1314.10
PAhhtype	Household type	nominal	Personal Details	529.51
PAhh_act	Employed persons in HH (16-64 years)	ordinal	Personal Details	1204.10
PAGroe	Height (self-declared in cm)	numeric	Personal Details	224.54
PAGewiB	Weight (self-declared in cm)	numeric	Personal Details	440.77
PAbmiB_k	BMI grouping (self-declaration)	nominal	Personal Details	551.04
PAbmiB_k2	Detailed BMI grouping (self-declaration)	nominal	Personal Details	563.60
PAueberB	Overweight according to WHO	binary	Personal Details	446.01
PAadiposB	Obesity according to WHO	binary	Personal Details	331.05
SDses_bild	Education/Training score	numeric	Sociodemographics	589.81
SDses_ber	Professional position score	numeric	Sociodemographics	305.28
SDses_score	Total Sociodemographic score	numeric	Sociodemographics	376.71
SDbild1D	Highest school leaving certificate	nominal	Sociodemographics	674.68
SDcasmin	CASMIN - educational groups	nominal	Sociodemographics	446.75
SDerwt0a	Living situation	nominal	Sociodemographics	1669.27
SDmainstat	Employment status	nominal	Sociodemographics	1580.24

Table 5.3: Overview of selected explanatory variables

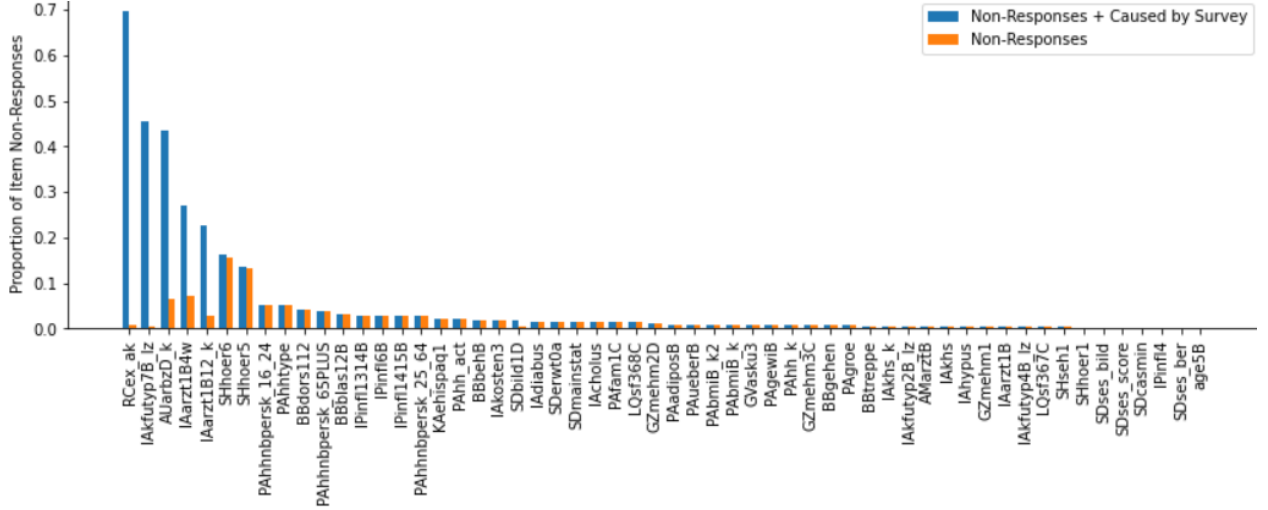


Figure 5.5: Item non-responses in explanatory variable realizations

example, the case if some question about smoking requires the individual to be a smoker in the first place. The reasons for the non-responses are highlighted by the colored bars in Figure 5.5. A deviation corresponds to missing values due to the survey structure. Taking the complete cases with respect to all non-responses would reduce the dataset to around 5% of the original number of instances, which is a severe information loss. Nevertheless, all algorithms require a complete input data matrix, whereby a minimal loss of information is desirable. This is solved with multivariate imputation of missing values.

5.2.3 Imputation of Missing Data

Let the realizations of the selected explanatory and target variables denote the dataset $\mathcal{D}$ of interest. Both the input matrix $\mathbf{X}$ resulting from the explanatory variable realizations and the target variable data have missing values. The former can be dealt with via imputation, while knowledge of all targets is a mandatory requirement. For the imputation, all instances of the input matrix are used. This subsection and its notation is based on van Buuren (2018).

Let the multivariate distribution of the hypothetically complete input dataset $\mathbf{X}$ be given by $p(\mathbf{X}|\boldsymbol{\omega})$ with parameters $\boldsymbol{\omega} \in \Omega_{\omega}$. However, missing values are contained in $\mathbf{X}$ that can be localized with an indicator matrix $\mathbf{R}$. For each element of $\mathbf{R}$ applies

$$r_{ij} = \begin{cases} 1 & , \text{if } x_{ij} \text{ is not observed,} \\ 0 & , \text{else} \end{cases} \quad (5.5)$$

and $\mathbf{X}$ splits into $(\mathbf{X}_{obs}, \mathbf{X}_{mis})$, i.e. observed and non-observed elements (van Buuren (2018), p. 35 f.). The distribution of the missing instances from equation 5.5 with the parameters $\boldsymbol{\psi} \in \Omega_{\psi}$ can depend on either $\mathbf{X}_{obs}$ and $\mathbf{X}_{mis}$. This relationship is called the missing data

mechanism and it is distinguishable three cases. Only depending to the parameters $p(\mathbf{R}|\boldsymbol{\psi})$, the missing values $\mathbf{R}$ are Missing Completely at Random. An additional dependency on the available data $p(\mathbf{R}|\mathbf{X}_{obs}, \boldsymbol{\psi})$ induces a Missing at Random mechanism. If the missing values depend on the underlying reason causing the missing values $p(\mathbf{R}|\mathbf{X}_{obs}, \mathbf{X}_{mis}, \boldsymbol{\psi})$, they are Missing Not at Random (Little & Rubin (2019), p. 13 f.). For each case, the given incomplete dataset can be described with $\mathbf{X}_{obs}$ and $\mathbf{R}$, where $p(\mathbf{X}_{obs}, \mathbf{R}|\boldsymbol{\omega}, \boldsymbol{\psi})$ corresponds to the associated joint density function. Contrary to the unknown parameters $\boldsymbol{\psi}$, the parameters $\boldsymbol{\omega}$ are estimable. In order to estimate $\boldsymbol{\omega}$ without knowledge of $\boldsymbol{\psi}$, the proportionality of the likelihood function of $\boldsymbol{\omega}$ and $\boldsymbol{\psi}$ to the joint density is used with

$$\ell_{full}(\boldsymbol{\omega}, \boldsymbol{\psi}|\mathbf{X}_{obs}, \mathbf{R}) \propto p(\mathbf{X}_{obs}, \mathbf{R}|\boldsymbol{\omega}, \boldsymbol{\psi}) \quad \text{which is} \quad \ell_{ign}(\boldsymbol{\omega}|\mathbf{X}_{obs}) \propto p(\mathbf{X}_{obs}|\boldsymbol{\omega}) \quad (5.6)$$

if $\boldsymbol{\psi}$ is ignorable (van Buuren (2018), p. 38). As shown by Little & Rubin (2019), $\boldsymbol{\psi}$ can be ignored in equation 5.6 if the missing values are Missing (Completely) at Random and if independence of the parameter spaces $\boldsymbol{\omega}$ and $\boldsymbol{\psi}$ holds, i.e. $\Omega_{\boldsymbol{\omega}, \boldsymbol{\psi}} = \Omega_{\boldsymbol{\omega}} \times \Omega_{\boldsymbol{\psi}}$. The Missing (Completely) at Random assumption is more important, since no dependence between $\boldsymbol{\omega}$ and $\boldsymbol{\psi}$ is apparent (Schafer (1997), p. 11). Under these conditions, $p(\mathbf{X}_{obs}, \mathbf{R}|\boldsymbol{\omega}, \boldsymbol{\psi})$ factorizes to

$$p(\mathbf{X}_{obs}|\boldsymbol{\omega})p(\mathbf{R}|\mathbf{X}_{obs}, \boldsymbol{\psi}) \propto \ell_{ign}(\boldsymbol{\omega}|\mathbf{X}_{obs})\ell(\boldsymbol{\psi}|\mathbf{X}_{obs}, \mathbf{R}), \quad (5.7)$$

where any likelihood ratio of the latter for $\boldsymbol{\omega}$ and an arbitrary $\boldsymbol{\omega}'$ is preserved with the omission of $\ell(\boldsymbol{\psi}|\mathbf{X}_{obs}, \mathbf{R})$ (Little & Rubin (2019), p. 133 ff.). The negligibly of $\boldsymbol{\psi}$ in equation 5.7 is a central result for the imputation of missing data, since

$$p(\mathbf{X}|\mathbf{X}_{obs}, \mathbf{R} = 1) = p(\mathbf{X}|\mathbf{X}_{obs}, \mathbf{R} = 0) \quad (5.8)$$

follows (van Buuren (2018), p. 39). Equation 5.8 implies that the distribution of the hypothetically complete data is identical to the distribution of the observed data and thus independent of $\mathbf{R}$ and $\boldsymbol{\psi}$. This allows for the identification of an imputation model on the observed data, that is then usable for the imputation of the missing values.

Entries that are missing due to the survey structure in Figure 5.5, are Missing Not at Random and multivariate imputation is not applicable. However, the reasons for the occurrence of the missing values are known. Possible solutions are the introduction of a new categorical level according to the survey structure or, alternatively, omitting the underlying variable. The latter option is chosen because the problem of missing values is severe for these cases in Figure 5.5. Moreover, variable relevance is likely to decrease with imputation in comparison to the available CS evaluation in Table 5.3. Accordingly, the variables IAarzt1B4w, IAarzt1B12_k, IAKfutyp7B_lz, AUarbzD_k and RCex_ak are removed.

Around 68% of the remaining input data instances are complete cases. For instances with missing values, several missing data points are possible. Let $\mathbf{X}_j$ be a data column vector of explanatory variable realisations, that possibly includes missing data. The multivariate imputation of $\mathbf{X}_j^{mis}$ is then based on $\mathbf{X}_{-j}$, which can contain missing values as well. In order to overcome this problem, Multiple Imputation by Chained Equations (MICE) is used as an imputation strategy (van Buuren & Groothuis-Oudshoorn (2011), p. 6 f.). MICE is based on the assumption that the multivariate distribution of the data $p(\mathbf{X}|\boldsymbol{\omega})$ can be determined by iterative draws from the conditional distributions $p(\mathbf{X}_1|\mathbf{X}_{-1}, \boldsymbol{\omega}_1), \dots, p(\mathbf{X}_k|\mathbf{X}_{-k}, \boldsymbol{\omega}_k)$. If $p(\mathbf{X}_j^{mis}|\mathbf{X}_j^{obs}, \mathbf{X}_{-j}, \mathbf{R})$ follows a multivariate imputation model, then MICE imputes $\mathbf{X}_j^{mis}$ $\forall j$ in the iterations $m = \{1, \dots, M\}$ via (van Buuren (2018), p. 120):

1. an initial imputation $\dot{\mathbf{X}}_j^{[0]}$ determined $\forall j$ by drawing with replacement from $\mathbf{X}_j^{obs}$,
2. updates of $\mathbf{X}_j^{[m]}$ $\forall j$ in iteration m with $\mathbf{X}_{-j}^{[m]} = (\mathbf{X}_1^{[m]}, \dots, \mathbf{X}_{j-1}^{[m]}, \mathbf{X}_{j+1}^{[m-1]}, \dots, \mathbf{X}_p^{[m-1]})$ as data columns that are used to build the multivariate imputation model,
3. one rotation in j which is one iteration in m , where $\forall j$ a draw from the parameter distribution $\dot{\boldsymbol{\omega}}_j^{[m]} \sim p(\boldsymbol{\omega}_j^{[m]}|\mathbf{X}_j^{obs}, \mathbf{X}_{-j}^{[m]}, \mathbf{R})$ leads to $\dot{\mathbf{X}}_j^{[m]} \sim p(\mathbf{X}_j^{mis}|\mathbf{X}_j^{obs}, \mathbf{X}_{-j}^{[m]}, \mathbf{R}, \dot{\boldsymbol{\omega}}_j^{[m]})$.

In the last step, the not directly determinable posterior predictive distribution of the multivariate imputation model is split into the posterior distribution of the observed data and the conditional predictive distribution of the missing data

$$p(\mathbf{X}_j^{mis}|\mathbf{X}_j^{obs}, \mathbf{X}_{-j}, \mathbf{R}) = \int_{\Omega_{\boldsymbol{\omega}}} p(\boldsymbol{\omega}_j|\mathbf{X}_j^{obs}, \mathbf{X}_{-j}, \mathbf{R})p(\mathbf{X}_j^{mis}|\mathbf{X}_j^{obs}, \mathbf{X}_{-j}, \mathbf{R}, \boldsymbol{\omega}_j)d\boldsymbol{\omega}_j \quad (5.9)$$

(van Buuren (2018), p. 42). Equation 5.9 makes MICE stochastic, which incorporates the uncertainty about the parameters due to the missing data by the initial draw of $\dot{\boldsymbol{\omega}}_j^{[m]}$ for the imputation $\dot{\mathbf{X}}_j^{[m]}$. The imputations of the data vectors with missing values is conditioned on all other data vectors via the second step. $\dot{\mathbf{X}}_j^{[0]}$ is successively updated with new values $\forall j$ and with each iteration m until the process converges to a stationary distribution of the parameters $\boldsymbol{\omega}_j$. In the literature, a minimum of 10 iterations is suggested and a detailed discussion of the conditions for convergence is given (van Buuren (2018), p. 120 ff.). Since the imputations in $m+1$ are only based on the values in m , the process corresponds to a Markov chain. Further, with clear and predefined transitions in the chain, the MICE algorithm is a variant of Gibbs sampling according to Geman & Geman (1984). With iterations set to 10 for the practical application, MICE forms the multivariate imputation framework.

Assuming the remaining missing values, which are not caused by the survey structure, to be Missing at Random, a suiting multivariate imputation algorithm has to be chosen. It is required to deal with categorical explanatory variables, which are the vast majority of Table 5.3. As described in section 3.2, Classification And Regression Trees (CART) are

adaptable to this. Since CART are non-parametric, their use implies an adaptation for MICE. After the initial imputation $\dot{\mathbf{X}}_j^{[0]} \forall j$, CART works in the MICE framework as described in the following process (Doove et al. (2014), p. 95):

1. in iteration m CART is iteratively fitted to $\mathbf{X}_{-j}^{obs}$ with $\mathbf{X}_j^{obs}$ as target, which creates trees with $Q_j^{[m]}$ leafs containing subsets of $\mathbf{X}_j^{obs}$ as possible donors for imputation,
2. for each missing value in $\mathbf{X}_j^{mis}$ determine the node by passing the respective entries in $\mathbf{X}_{-j}^{mis}$ through the fitted tree and draw $\dot{\mathbf{X}}_j^{mis}$ randomly from the node candidates,
3. update the former imputation by rotation in $j = \{1, 2, \dots, k\}$ for one iteration m .

CART emulates the stochastic nature of MICE by random draws from a deterministic solution. Applying MICE with CART and $M = 10$ iterations yields a complete data matrix of the dimension $24,016 \times 51$. However, knowledge about the targets is required for algorithm training and evaluation. The complete cases of the target variable data reduce the usable instances. This leaves a complete dataset $\mathcal{D}_{imp}$ with 22,185 instances.

5.2.4 Data Splits, Encoding and Scaling

Further preprocessing steps are required for the performance evaluation and selection of the probabilistic base algorithm from chapter 3 as well as the performance evaluation of the multi-label approaches from chapter 4. In line with equation 2.16 and equation 2.17, data splits in a training, validation and test sets have to be conducted to get generalization error estimates. This allows the evaluation of the predictive performance of algorithms and approaches on untrained instances. Further, the majority of algorithms are unable to process the nominal variables in Table 5.3. An encoding to a numerical representation is needed. An approach that captures prior knowledge about the targets and the marginal dependency at the category levels is utilized. Lastly, some algorithms in chapter 3 showed that a common scale of the explanatory variables is beneficial. At the same time, no reason for harm to the predictive performance of the other tree based algorithms is apparent. Accordingly, explanatory variables are scaled to the intervall $[0, 1]$.

Splits of the completed dataset $\mathcal{D}_{imp}$ into training, evaluation and test sets by resampling are used to introduce experimental setups for predictive performance evaluation. These experimental setups for the evaluation of the data preprocessing, the probabilistic base algorithm and the multi-label approaches are discussed in detail in section 5.4. Any split is required to result in mutually exclusive data partitions, which implies draws of instances without replacement from the dataset. Further, the empirical distributions of the target variables are to be preserved to get representative sub-samples for any training, validation and test set. This is supposed to ensure a representative train- and test-ability of any

algorithm or approach. Neglecting the two-stage stratified cluster sampling in equation 5.2, since no population level inference is required, random samples of instances are used for splits. Moreover, the amount of data allocation to the sets resulting from splits is of interest. As discussed in section 2.1, discriminative algorithms benefit from an increasing amount of training instances, which results at the same time in higher computational costs. However, larger amounts of test instances provide increased certainty about the evaluation results. Following heuristics for a simple test set strategy, the dataset $\mathcal{D}_{imp}$ can be split randomly by assigning 70%, i.e. 17,748 instances, to $\mathcal{D}_{train}$ and by assigning the remaining 30%, i.e. 4,437 instances, to $\mathcal{D}^{test}$ such that $\mathcal{D}_{test} = \mathcal{D}_{imp} \setminus \mathcal{D}_{train}$ is an untrained holdout set. Note that cross-validation with random sampling is a more sophisticated version of this. Regardless of the underlying dataset as well as the number of folds, each dataset instance is considered once for evaluation during cross-validation. Here, Fushiki (2011) finds that increasing the size of the folds results in a less biased estimate of the generalization error. However, the burdens for an increase in folds are the associated computational complexity. This is why, unless otherwise stated, a 5-fold cross-validation is considered whenever needed in the computational part of this thesis.

The majority of the variables underlying $\mathcal{D}_{imp}$ are of nominal scale and have to be encoded in a numerical representation. A straightforward approach is One-Hot Encoding (OHE), which generates a dummy variable for each category level of a nominal variable. However, the realizations of the targets in $\mathcal{D}_{train}$ can be advantageously incorporated in the encoding as for example shown in de la Bourdonnaye & Daniel (2021). One of those encoding methods is the Weight-of-Evidence (WoE), which stems from credit scoring. For each nominal variable, that is indicated with the respective index j by X_j , and per category c of C category levels, the WoE with the target variable realizations of Y_l in the column vector $\mathbf{Y}_l$ is given by

$$\text{WoE}_{jcl} = \log\left(\frac{P(X_j = c|Y_l = 0)}{P(X_j = c|Y_l = 1)}\right) \quad \text{with} \quad \hat{P}(X_j = c|Y_l = 0) = \frac{\#\mathcal{D}_{train}\{\mathbf{X}_j = c, \mathbf{Y}_l = 0\} + \alpha}{\#\mathcal{D}_{train}\{\mathbf{Y}_l = 0\} + 2\alpha}. \quad (5.10)$$

Transforming all C categories with equation 5.10 and with respect to target l yields $\mathbf{X}_{jl}^{enc}$. The encoding of one input matrix column follows from substitution of $\mathbf{X}_j$ with $\mathbf{X}_{jl}^{enc} \forall l$. The estimate is based on $\mathcal{D}_{train}$, where $\#$ implies the count of instances for $\mathbf{X}_j^{train}$ belonging to the respective category c and the respective class of target $\mathbf{Y}_l$. The WoE is given as proportion of the respective category in the negative as well as positive class by the total occurrence of the classes. It assesses whether the ratio between the classes changes at the categorical level. In order to prevent a division by zero in equation 5.10, Laplace smoothing with a smoothing parameter $\alpha = 1$ is used as pseudo counts (Ng & Jordan (2001), p. 2). Note that $\mathcal{D}_{test}$ is not incorporated in the estimation, since the targets are considered unknown.

However, the WoE estimates can be used to transform the respective nominal variable values in $\mathbf{X}_{test}$. Here, the WoE is set to zero if a category level appears in $\mathbf{X}_{test}$ but not in $\mathbf{X}_{train}$. Transforming each nominal variable with respect to all labels l , encodes prior and cross target knowledge. This enables algorithms to incorporate marginal dependency at the category level in the predictions. Encoding the imputed input matrix data based on a split in $\mathbf{X}_{train}$ and $\mathbf{X}_{test}$ increases the number of columns in the matrices from 51 to 169.

In order to account for the benefits of scaling, as shown theoretically in the case of the regularized LR from section 3.1 and the MLP as introduced in section 3.3, a Min-Max Scaler is used. It normalizes all ordinal, numeric and WoE encoded nominal input data displayed in Table 5.3 to the interval $[0, 1]$. Let $j = \{1, \dots, 169\}$ take the index values of the relevant variables, then scaling is done $\forall i$ in column j by

$$x_{ij}^{scaled} = \frac{x_{ij} - \mathbf{X}_j^{min}}{\mathbf{X}_j^{max} - \mathbf{X}_j^{min}}. \quad (5.11)$$

In equation 5.11, $\mathbf{X}_j^{max}$ is the maximum value and $\mathbf{X}_j^{min}$ the minimum value of column j in the known instances of $\mathbf{X}_{train}$. Accordingly, all variable realizations that show values outside $[0, 1]$ are transformed and substituted in $\mathbf{X}_{train}$ and $\mathbf{X}_{test}$. The transformation of $\mathbf{X}_j^{test}$ can cause values outside $[0, 1]$, e.g. by values exceeding the respective maximum in $\mathbf{X}_j^{train}$. However, those values are truncated at the interval bounds. For tree based algorithms from section 3.2, no harmful impact of scaling on the predictive performance is apparent. In fact, the split results of the underlying trees should remain the same due to scale invariance.

Note that $\mathbf{X}_{train}$ and $\mathbf{X}_{test}$, underlying the split in $\mathcal{D}_{train}$ and $\mathcal{D}_{test}$, represent the preprocessed data in the following, i.e. after imputation, encoding and scaling. All algorithms from chapter 3 and approaches from chapter 4 can be fitted to $\mathcal{D}_{train}$ and evaluated with $\mathcal{D}_{test}$.

5.3 Performance Metrics

Having the MBC loss in equation 2.14 as criterion to be optimized for the identification of the algorithms and approaches via equation 2.16, other metrics can be used to assess the predictive performance based on $\mathcal{D}_{test}$. This is done by evaluation of the predictions with respect to the ground truth of the test data targets, e.g. see Figure 5.7. Selecting more than one evaluation metric is desirable, since each one only captures specific information and varying aspects about the properties of the predictions are of interest. However, the following evaluation measures have a direct evaluation of the conditional probability distributions in common. Further, all performance metrics are aggregated across all labels, such that a label-wise decomposition is possible. This is conducted in section 6.1 to gain insights about performance for each label.

A evaluation metric closely related to the MBC loss and equation 2.17, is the Brier Score (BS) by Brier (1950). It is a strictly proper scoring rule intended to confirm the minimization of the MBC loss for the multi-label approaches (Gneiting & Raftery (2007), p. 365).

$$BS_{Avg} = \frac{1}{L} \sum_{l=1}^L \frac{1}{|\mathcal{D}_{test}|} \sum_{(\mathbf{x}_i, y_{il}) \in \mathcal{D}_{test}} \left(\hat{h}_l(\mathbf{x}_i) - y_{il} \right)^2 \quad (5.12)$$

is used, which is the average of the label-wise BS. Relying on the squared distance of the probabilistic prediction and the true class, it can be seen as the probabilistic equivalent to the mean squared error used in the regression framework. Note that equation 5.12 does not penalize deviations between the probabilistic prediction and the true class as in equation 2.14. The average BS takes values in $[0, 1]$ where lower values indicate better performance.

Predicted	Actual	
	1	0
1	TP_l	FP_l
0	FN_l	TN_l

Table 5.4: Confusion matrix for label l

In order to introduce further performance metrics, a threshold value $\tau \in (0, 1)$ is considered that dichotomies the values of a predicted conditional probability distribution into $\{0, 1\}$, i.e. the negative and the positive class. For label l , the predicted positive class follows if $h_l(\mathbf{x}) \geq \tau$ and the negative class else. More generally, this can be referred to as classification. Four different outcomes between a classification and the true outcome are possible and depicted in Table 5.4. In this so-called confusion matrix, TP_l counts the true positive outcomes where the classified positive class match the actual positive outcome. FP_l counts the false positive outcomes where the classified positive class does not match the actual negative outcome. The same logic applies to TN_l and FN_l with respect to the negative class. The Positive Predictive Value (PPV), the True Positive Rate (TPR) and the False Positive Rate (FPR) can be derived from Table 5.4 as

$$PPV_l = \frac{TP_l}{TP_l + FP_l}, \quad TPR_l = \frac{TP_l}{TP_l + FN_l} \quad \text{and} \quad FPR_l = \frac{FP_l}{FP_l + TN_l}, \quad (5.13)$$

which take $\forall l$ and $\forall \tau$ values in the interval $[0, 1]$. In the equations of 5.13, the PPV_l describes the proportion of matched true positives per all positively predicted instances. The TPR_l describes the proportion of matched true positives per all actually possible positive outcomes and the FPR_l the proportion of incorrectly identified outcomes as positives per all actually possible negative class outcomes. Note that the dependence on the threshold τ implies that varying realizations in Table 5.4 and hence for the values of PPV_l , TPR_l and FPR_l are

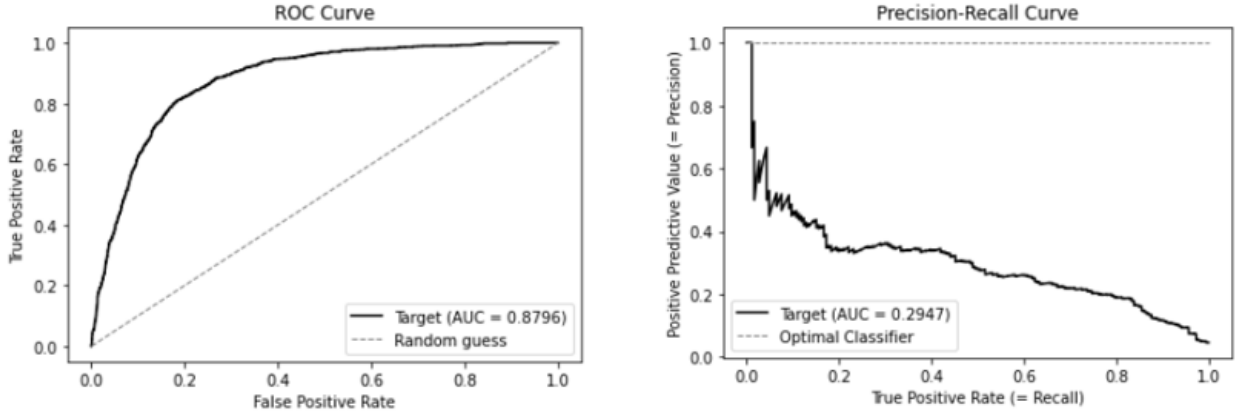


Figure 5.6: Area under the ROC and PRC

possible by threshold moving in $(0, 1)$. Any non-perfect situation results from a predicted conditional probability distribution, that is not perfectly discriminative in its ranking such that there does not exist a τ that causes the classification to match the actual class outcome for all test instances. Here, a trade-off between TPR_l and FPR_l as well as TPR_l and PPV_l exists along the possible values of τ . Taking a substantial amount of candidate values for τ and calculating the respective values for FPR_l , TPR_l and PPV_l from Table 5.4, one can obtain curves as interpolated graphical representations. This is illustrated for an arbitrary target in Figure 5.6, where the Receiver Operating Characteristic (ROC) is given by the TPR and FPR and the Precision-Recall Curve (PRC) by the PPV and TPR trade-off.

The theoretic ROC on the left hand side of Figure 5.6 depicts the discriminative capability of a multi-label approach in terms of the ranking of the probabilistic predictions. An optimal approach yields a solution in the upper left corner, which implies that all actual positives and negative instances are identified in the correct order at every threshold τ . In that case, the lowest probability prediction for the positive class is higher than the highest probability prediction for the negative class. The diagonal line defines random guessing where the rates remain equivalent across τ such that $TPR_l = FPR_l \forall \tau$.

Replacing the FPR_l with PPV_l , also called Precision, and focusing on the trade-off with TPR_l , also called Recall, the right hand side of Figure 5.6 depicts a theoretic PRC. An optimal approach yields a solution in the upper right corner. Note the sharper focus on the positive class. While the TPR_l increases with falling τ and inflating positive class classifications, PPV_l accounts more heavily for emerging false positives. A large value for PPV_l is attained if the multi-label approach precisely predicts the positive class across τ .

In order to reduce both curves in Figure 5.6 to metrics, one can compute the area under the respective curve. This is approximately done with the help of the trapezoidal rule as, for example, described in Tallarida & Murray (1987). Both $AUROC$ and $AUPRC$ take values in the interval $[0, 1]$ with an optimal multi-label approach having a value of 1. Nevertheless,

the multi-label framework requires a combination of the calculations with respect to each label l into one metric. This implies either a calculation across all labels or a combination of the label-wise metrics (Herrera et al. (2016), p. 59). The latter is chosen to prevent a dilution of rare labels which neglects label imbalance. A simple strategy of averaging across labels is considered by

$$Metric_{Avg} = \frac{1}{L} \sum_{l=1}^L Metric_l(TP_l, FP_l, FN_l, TN_l) . \quad (5.14)$$

Calculating $AUROC$ and $AUPRC$ for each label and averaging the result in equation 5.14, yields $AUROC_{Avg}$ and $AUPRC_{Avg}$ as further performance metrics of interest. Note that the averaging strategy allows for an easy decomposition in the label-wise metrics.

So far performance metrics focused on the predictive performance of the multi-label approaches. This is accompanied by the performance in terms of the computational complexity. Accordingly, the time for fitting and producing predictions for each multi-label approach is captured in seconds and added as complexity metrics.

5.4 Approach Optimization and Experimental Setups

This section follows the theoretic foundations in section 2.3 as well as the considerations for splits in subsection 5.2.4. The objective is to find optimized specifications of the probabilistic base algorithms and the multi-label approaches based on $\mathcal{D}_{train}$. Moreover, experimental setups are introduced such that a evaluation can be conducted on $\mathcal{D}_{test}$ with the performance metrics from section 5.3. Note that three possible sources of improvement in predictive performance can be identified and evaluated. Firstly, the data preprocessing in Figure 5.3 that has a general influence on the predictive performance of all algorithms and multi-label approaches. Secondly, the most multi-label approaches depend on an underlying probabilistic base algorithms which can be optimized and chosen. Thirdly and of central interest, the evaluation of the resulting multi-label approaches them-self. The evaluation of every possible combination of general influence factors, probabilistic base algorithm and multi-label approaches is infeasible. Accordingly, three experimental setups are used to evaluate each source of potential improvement in predictive performance separately. Note that the experimental setups for the data preprocessing and the probabilistic base algorithms are kept simple since those represent auxiliary steps to the major and more sophisticated comparison of the multi-label approaches.

For the evaluation of the data preprocessing in Figure 5.3, no optimization of algorithms and approaches is performed. The simple holdout strategy detailed in subsection 5.2.4 is used. This implies a random and mutual exclusive split of the imputed data, where 70% is

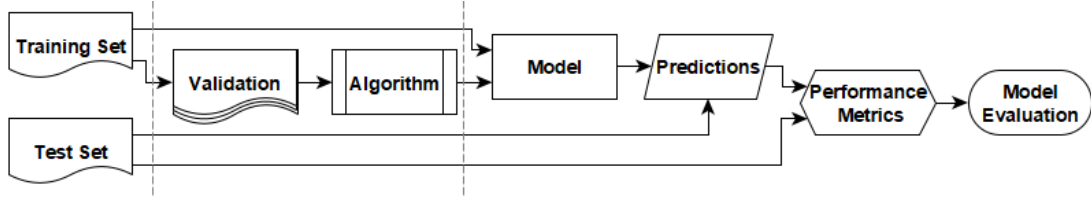


Figure 5.7: Evaluation of data preprocessing and probabilistic base algorithms

allocated to the training set $\mathcal{D}_{train}$ and 30% is allocated to the test set $\mathcal{D}_{test}$. This is followed by the encoding and scaling as described in subsection 5.2.4. The baseline combination of BR and a non-regularized LR is used to obtain test set predictions. Following this, the evaluation is conducted with the metrics from section 5.3 and according to Figure 5.7 by neglectation of the highlighted validation part. The imputation and encoding method used in section 5.2 are benchmarked against simple alternatives, i.e. the imputation with the most frequently observed value for each column of $\mathbf{X}_{train}$ and OHE. Further, an evaluation with and without scaling is conducted. An evaluation of variable selection techniques is considered out of scope.

The evaluation of the probabilistic base algorithms requires an optimization of the respective hyperparameters as outlined in chapter 3. This is done with BR from equation 4.1 for applicability to the multi-label framework. In line with minimization of MBC from equation 2.14 as well as the minimization of the expected generalization error as described in equation 2.16, the simple holdout strategy is extended by a 5-fold cross-validation on $\mathcal{D}_{train}$ as depicted in Figure 5.7. The hyperparameter specification that minimizes the MBC is identified for each probabilistic base algorithm. Since multiple hyperparameters result in a grid of candidate combinations that increases complexity quadratically by the grid dimensionality, this solution might represent a local minimum due to the computational complexity attached. Fitting this optimized specification of the probabilistic base algorithms together with BR to the training data $\mathcal{D}_{train}$ yields the final model that is used for evaluation. The process is illustrated with the complete workflow in Figure 5.7, that results in the evaluation on $\mathcal{D}_{test}$ with the metrics from section 5.3. Here, the evaluation is intended to confirm the optimization of the expected generalization error for the respective model.

The optimization of the multi-label approaches is conducted in a more approach specific manner. For the Problem Transformation approaches from section 4.1 and the homogeneous ensemble approaches from section 4.3, the best suiting probabilistic base algorithm is selected as basis according to their evaluation. Afterwards, the heuristic default specifications outlined in chapter 4 are considered. In order to determine the chain order in CC and FCD, the predictive performance of all permutations is evaluated with CC based on $\mathcal{D}_{test}$ in the simple holdout strategy. The Algorithm Adaptation approaches from section 4.2 need a more comprehensive optimization. MLRF is optimized similar to a probabilistic base algorithm using a 5-fold cross-validation. For DNN the simple holdout strategy is utilized together with early stopping.

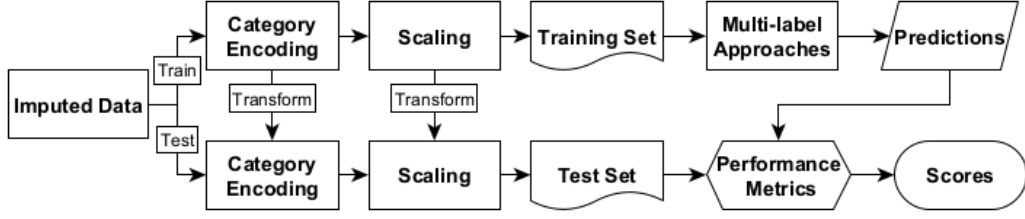


Figure 5.8: One iteration and fold in the 5×2 -fold cross-validation

The parameters of DNN are sequentially and manually optimized, because of the flexible structure. The heterogenous ensemble approaches from section 4.3 utilize combinations of varying probabilistic base algorithms as well as several multi-label approaches. HSE uses all probabilistic base algorithms and HBE the two most suiting probabilistic base algorithms in combination with BR. In addition to the approaches considered for HBE, HBE_{opt} incorporates the two most promising multi-label approaches according to the evaluation.

Having identified the specifications of the multi-label approaches, a more sophisticated evaluation strategy is chosen for the multi-label approach comparison. So far the uncertainty due to random splits in training and test set is neglected. However, the associated variance in the results is of central interest to infer significant differences between the approaches. Dietterich (1998) introduces the 5×2 fold cross-validation accompanied by a paired t-test for statistical significance in the evaluation result differences between approaches. A single evaluation step of all multi-label approaches is illustrated in Figure 5.8. The dataset $\mathcal{D}$ is randomly split in equal sized partitions generating 2 folds and hence both partitions are once considered as $\mathcal{D}_{train}$ as well as $\mathcal{D}_{test}$. The folds yield two results of the performance metrics in section 5.3, that are averaged for each multi-label approach. These fold averages of the multi-label approaches are generated for 5 random splits in total. A further average over the 5 iterations is generated, which results in the point estimate of the expected performance of the multi-label approaches. Based on this, the 5×2 cross-validation t-test is conducted. Let A and B be two multi-label approaches that are to be tested for differences in performance. Further, let e represent an evaluation metric of interest, then the two possible differences of the approaches in evaluation metric results on the two folds are given by $d_1 = e_A^{(1)} - e_B^{(1)}$ and $d_2 = e_A^{(2)} - e_B^{(2)}$ via the two fold alternation of training and test set. The variance in this differences at iteration $m = \{1, 2, \dots, 5\}$ is given by

$$s_i^2 = \left(d_1^{[m]} - \bar{d}^{[m]}\right)^2 + \left(d_2^{[m]} - \bar{d}^{[m]}\right)^2 \quad \text{where} \quad \bar{d}^{[m]} = \frac{\left(d_1^{[m]} + d_2^{[m]}\right)}{2}. \quad (5.15)$$

Let $d_1^{[1]}$ be the metric difference of the multi-label approaches for the first fold of the first

iteration. With the results from equation 5.15 follows the test statistic

$$t_{score} = \frac{d_1^{[1]}}{\sqrt{\frac{1}{5} \sum_{i=1}^5 s_i^2}} \stackrel{H_0}{\sim} t_5 . \quad (5.16)$$

It is assumed to follow approximately a t-distribution t with 5 degrees of freedom in equation 5.16 (Dietterich (1998), p. 10 f.). A two sided-test is conducted, hence the multi-label approaches A and B are assumed to have equal performance under the null hypothesis H_0 . H_0 is rejected if the absolute value of the test statistic, for a given significance level α , is larger then the respective critical value of the t-distribution. Equivalently, the P-value of the test statistic result can be used for direct comparison with α . Moreover, Dietterich (1998) show that the paired t-test based on the 5×2 cross-validation, is more powerful then paired t-tests based on repeated random splitting or cross-validation. Note that there is additional uncertainty in Figure 5.8 due to the encoding and scaling that depends on the split. The test is not corrected for this. The test statistic results can be considered to be rather conservative, such that test statistic values might be too small.

6 Results

For the evaluation in the experimental setups, the MBC loss from equation 2.14, which underlies optimization by validation, is reused as in 2.17 to assess predictive performance. It is reported together with the introduced performance metrics BS_{Avg} , $AUROC_{Avg}$, $AUPRC_{Avg}$ and the computational time in seconds. Note that lower values indicate better performance for the MBC and BS_{Avg} , whereas higher values are desirable for $AUROC_{Avg}$ and $AUPRC_{Avg}$. With that, the evaluation of the data preprocessing, the probabilistic base algorithms as well as the multi-label approaches is conducted in the respective optimizations and experimental setups outlined in section 5.4. The evaluation results are presented in four result tables. The first two shows the auxiliary results of the data preprocessing and the optimized probabilistic base algorithms that are based on the same simple holdout evaluation strategy. The third and fourth table resembles the major evaluation part. Here, the comparison of the multi-label approaches as well as the tests for significance in the performance differences based on the 5×2 cross-validation are presented. This is followed by a discussion of all evaluation results.

6.1 Evaluation

The data preprocessing methods in section 5.2 are compared using simple benchmarks. Those are denoted by Simple Imputation, which uses the most frequently observed value of the observed data per column, and OHE, which generates a dummy column for each category level. The varying scenarios depicted in Table 6.1 result from the available data preprocessing

Scenario:	MBC	BS_{Avg}	$AUROC_{Avg}$	$AUPRC_{Avg}$	$Time$
1. Simple + OHE	0.9317	0.0834	0.8605	0.4301	2.82
2. CART + OHE	0.9088	0.0806	0.8706	0.4519	1.88
3. CART + WoE	0.9065	0.0802	0.8720	0.4570	8.81
4. CART + WoE + Scaling	0.9039	0.0800	0.8726	0.4574	3.91

Table 6.1: Comparison of different data preprocessing scenarios

options. Their evaluation is based on a non-regularized BR-LR baseline. The initial scenario utilizes the simple benchmark preprocessing options. In the second scenario, the imputation method is changed to MICE with CART. The third scenario alters the encoding method to WoE. In the last scenario, the Min-Max Scaler is additionally considered. The holdout strategy underlying Table 6.1 provides evidence that the use of the data preprocessing

method from section 5.2 is beneficial with respect to the predictive performance metrics and in comparison to the simple benchmark methods. All evaluation metrics improve as the methodology is successively switched away from the benchmarks. The use of the multivariate imputation with CART in scenario two, achieves the highest performance increase. Note that the WoE increases the dimensionality more than OHE, resulting in higher computational costs in the third scenario.

For the evaluation of the probabilistic base algorithm, the same holdout strategy is used. However, cross-validation on the training data identifies optimized algorithm specifications. The inverse of the regularization strength $\frac{1}{\lambda}$, required by the LR implementation, is found to be 0.625. The RF trees are build on bootstrapped samples consisting of 35% of the original training data size, with a maximum tree depth of 6 and 40% of the input matrix columns considered at each split. The ensemble size is heuristically increased to 250. GB is build with 150 boosting iterations, trees of maximum depth 2 and a learning rate η of 0.08. The learning rate of MLP is identified at 0.0013 with a hidden layer size of 110. The results of this specifications on the simple holdout are given in Table 6.2. The only difference

Probabilistic Algorithm	MBC	BS_{Avg}	$AUROC_{Avg}$	$AUPRC_{Avg}$	$Time$
BR - LR	0.9024	0.0799	0.8736	0.4602	14.27
BR - RF	0.9083	0.0810	0.8683	0.4444	41.68
BR - GB	0.9035	0.0801	0.8720	0.4549	7.84
BR - MLP	0.9246	0.0822	0.8559	0.4290	12.81

Table 6.2: Comparison of optimized probabilistic base algorithms

between BR-LR and the fourth scenario in Table 6.1 is the regularization of LR. Here, the regularization shows some improvements on all evaluation metrics at the cost of a higher computational time. Overall Table 6.2 indicates that BR-LR and BR-GB are the most suiting probability base algorithms with BR-LR being slightly ahead in all evaluation metrics except the computational time. The computational time of BR-RF is the highest due to the large ensemble size. BR-MLP performs remarkably worse than the other algorithms. Considering the low level of hyperparameter optimization required and the generic applicability to altering settings, LR is chosen in favour of GB as probabilistic base algorithm for the multi-label approaches. However, BR-GB is included in the evaluation of the multi-label approaches. Note that the evaluation of the probabilistic base algorithm, like the evaluation of the data preprocessing methods, depends only on the single split. This does not represent a throughout evaluation of performance differences. However, the experiments are considered to be sufficient to be used as auxiliary and intermediate results.

The evaluation results of $AUROC_{Avg}$ and $AUPRC_{Avg}$ for BR-LR can be decomposed label-wise to assess the label specific performances. Figure 6.1 shows the decomposed ROC

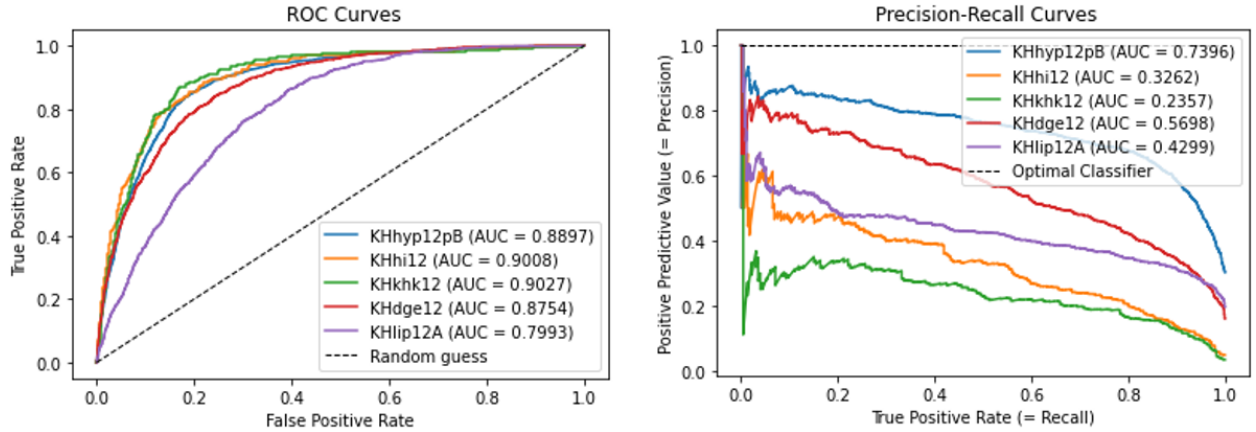


Figure 6.1: Label-wise decomposition of $AUROC_{Avg}$ and $AUPRC_{Avg}$

on the left hand side and the decomposed PRC on the right hand side. Each color represents one of the underlying diseases as detailed in Table 5.2. The respective values for the area under the curve are given in the legends. Except for **KHlip12A**, all disease targets show a good $AUROC$ value. A specific reason for this is not apparent. For the $AUPRC$, several disease target show low values. Comparing this to the relative frequency of the positive class in Table 5.2 makes apparent, that the precision across the possible thresholds is low for highly imbalanced targets. For **KHkhk12**, only 3.78% of the data belong to the positive class. For **KHhi12**, the proportion is 4.39%. There are too few data instances belonging to the positive class such that the discriminative boundaries of the multi-label approaches become imprecise. One can conclude that imbalanced data makes an exact identification of the positive class in general challenging (Herrera et al. (2016), p. 133 ff.).

With the identification of LR as suiting probabilistic base algorithm, several multi-label approaches are considered in the heuristic specification as outlined in chapter 4. Nevertheless, chain orders, Algorithm Adaptation approaches and heterogeneous ensemble approaches are identified. For the first two, the simple holdout split is used again. The order of the targets in the chain of CC and FCD is found to match the CS order in Table 5.2 from highest to lowest. For MLRF trees are build on bootstrapped samples having 85% of the original training data size and with a maximum depth of 12 where 30% of the data columns are considered at each split. The ensemble size is increased to 250. DNN has two hidden layers of size 110 with $\frac{1}{3}$ of the connecting weights dropped out. The learning rate η is set to 0.05. HBE is applied to the predictions underlying the results of BR-LR and BR-GB from Table 6.3. Additionally, HBE_{opt} considers the predictions of FCD and HSE in the ensemble of four approaches. Note that the experimental setup for the evaluation of the multi-label approaches is different and incomparable to Table 6.1 as well as Table 6.2. Having the 5×2 cross-validation of the whole dataset underlying, the results in Table 6.3 represent expected performance metrics. With respect to BR-LR as benchmark, the Algorithm Adaptation approaches show

Multi-label Approach	MBC	BS_{Avg}	$AUROC_{Avg}$	$AUPRC_{Avg}$	$Time$
BR-LR	0.9171	0.0816	0.8702	0.4532	2.81
BR-GB	0.9144	0.0819	0.8700	0.4525	4.11
LP	0.9245	0.0828	0.8660	0.4386	29.75
CC	0.9186	0.0817	0.8701	0.4538	2.75
FCD	0.9161	0.0816	0.8709	0.4539	38.49
MLRF	0.9269	0.0834	0.8626	0.4408	13.79
DNN	0.9187	0.0821	0.8692	0.4497	27.36
EBR	0.9166	0.0816	0.8701	0.4540	14.27
BRS	0.9189	0.0819	0.8691	0.4473	37.71
ECC	0.9198	0.0819	0.8696	0.4541	14.41
HSE	0.9346	0.0827	0.8712	0.4576	146.91
HBE	0.9126	0.0813	0.8716	0.4574	-
HBE _{opt}	0.9147	0.0814	0.8721	0.4582	-

Table 6.3: Expected performance of multi-label approaches via 5×2 cross-validation

no improvement in any performance metric. Upon the Problem Transformation approaches, only FCD is improving in all evaluation metrics at the expense of higher computational costs. The ensemble approaches EBR, BRS and HSE display improvements in $AUROC_{Avg}$ and $AUPRC_{Avg}$. For the latter, which additionally utilizes differences in varying probabilistic base algorithms, the improvement is more substantial. However, the performance with respect to MBC and BS_{Avg} gets worse and computational costs are much higher. HBE and HBE_{opt} show an improvement in all evaluation metrics with HBE_{opt} being ahead in $AUROC_{Avg}$ as well as $AUPRC_{Avg}$ and HBE being ahead in BS_{Avg} . The computational cost of all underlying approaches is required and hence the largest of all approaches, but not directly determined here. In total, the performance of both FCD as well as HBE_{opt} is promising.

In order to assess the statistical significance of the improvements, the 5×2 paired t-test is conducted as described in section 5.4. It tests for the differences in the evaluation metrics of the benchmark approach BR-LR and FCD as well as HBE_{opt}. The resulting P-values, given the t-statistic in equation 5.16 and the t-distribution with 5 degrees of freedom, are displayed in Table 6.4. This is accompanied with the standard deviation estimate for the differences due to the variation in the splits, that captures the attached uncertainty. For a significance level of $\alpha = 5\%$, Table 6.4 shows that the differences in the evaluation metrics of BR-LR and FCD are not statistically significant. The assumption of equal performance under H_0 can not be rejected. However, the differences in BS_{Avg} and $AUROC_{Avg}$ for BR-LR and HBE_{opt} are statically significant at $\alpha = 5\%$. Still, the differences in $AUPRC_{Avg}$ remain insignificant. This is mainly driven by the higher uncertainty in $AUPRC_{Avg}$ as indicated by the value of the standard deviation in Table 6.4. In line with the decomposition in Figure 5.8, an obvious explanation for that is the imbalance of the labels that causes increased uncertainty in the

$H_1 : A \neq B$ with $A = \text{BR-LR}$	BS_{Avg}	$AUROC_{Avg}$	$AUPRC_{Avg}$
P-values for $B = \text{FCD}$	0.3605 (.00003)	0.1251 (.00025)	0.6051 (.00114)
P-values for $B = \text{HBE}_{opt}$	0.0074 (.00004)	0.0218 (.00035)	0.0845 (.00127)

Table 6.4: P-values of 5×2 paired t-test with standard deviation in parenthesis

data split. The allocation of single positive class instances is of higher relevance and thus causes the $AUPRC_{Avg}$ results to fluctuate more.

6.2 Discussion

In this thesis, several steps affecting the prediction performance are considered, with the comparison of the candidate multi-label approaches being of central interest. Those steps are illustrated on the left hand side of Figure 6.2. The selected data preprocessing methodology for imputation, encoding and scaling shows to have a positive influence on the predictive performance in terms of the metrics considered in section 6.1 and with respect to more simple alternatives. This result has to be interpreted with caution, since it is based on metrics calculated on a one split holdout strategy. However, it is theoretically backed by section 5.2 and the overall improvement identified in Table 6.1 is noticeably large in comparison to the improvements stemming from the multi-label approaches in Table 6.3. Following this, there is arguably further potential for improvements in predictive performance with other and more sophisticated data preprocessing steps. Those have the potential to be more beneficial than the consideration of label dependencies via the multi-label approaches in Table 6.3. Anyway, a more comprehensive analysis of the data preprocessing steps can be advantageous, either to gain more confidence in the evaluation results and to assess further preprocessing options. This also includes variable selection in Figure 6.2 that is initially fixed and not part of the evaluation. Here, Kashef et al. (2018) provide evidence that predictive performance can be improved by the search for the most suiting variable selection technique. Due to the same evaluation setup, a comparison of the last scenario in Table 6.1 and Table 6.2 is possible. The optimization of LR shows minor benefits and the other probabilistic base algorithms yield no improvement in predictive performance. Due to the uncertainty not considered for this result, the two best base algorithms LR and GB are used together with BR in the comparison of multi-label approaches. This comparison is the central building block of the evaluation and highlighted in section 6.1. The more sophisticated experimental setup of the 5×2 cross-validation yields expected metrics results for the predictive performance in Table 6.3. Here, neither BR-LR nor BR-GB are clearly superior. Upon the approaches that utilize only the dependency between the labels, FCD shows the greatest improvement in predictive performance. However, the 5×2 paired t-test result in Table 6.4 identifies the

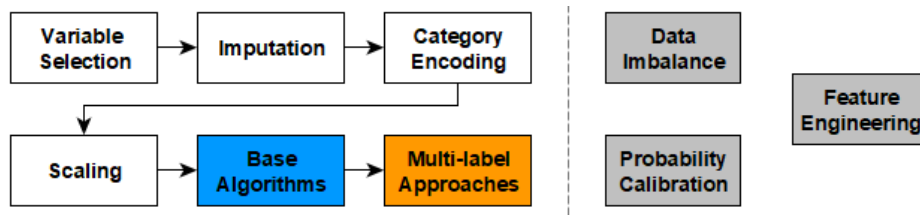


Figure 6.2: Evaluated and observed topics for improvement of predictions

difference to the benchmark approach BR-LR as statistically insignificant for $\alpha = 5\%$. By an additional utilization of several probabilistic base algorithms in heterogeneous ensembles, the probabilistic performance increases further. Taking advantage of the variation and individual strengths of the probabilistic base algorithms in HBE_{opt} , shows to improve the predictive performance across all evaluation metrics even more than the use of label dependency. This provides a statistically significant improvement over BR-LR for two performance metrics, as shown by the test results in Table 6.4. Apart from label dependency, the use of heterogeneity in probabilistic base algorithms and multi-label approaches is further source of improvement for predictive performance. However, issues raised and observed, but not addressed in this thesis, are illustrated on the right hand side of Figure 6.2. Table 5.2 showed imbalance in the disease labels, which leads to a deteriorated identification of the minority class by the multi-label approaches in Figure 5.8. This increases the uncertainty in the evaluation results of $AUPRC_{Avg}$. Feature Engineering was mentioned in subsection 5.2.2, but not pursued further due to the limited scope. Following the discussion of probabilistic classifiers in section 2.1, calibration is a possible next step following the results in Table 6.3 either to ensure the interpretation as probability or to improve the predictive performance with respect to strictly proper scoring rules even further.

7 Conclusion and Outlook

This thesis analyses the probabilistic prediction quality for diseases in terms of discrimination, in a multi-label framework for multiple disease targets and based on the German Health Update survey data. Mandatory preprocessing steps that deal with the peculiarities of the survey data are conducted within this multi-label framework. Those include target and explanatory variable selection, imputation of missing values, encoding of categorical variables as well as scaling. While variable selection is not evaluated, the remaining preprocessing steps are benchmarked against simple alternatives to assess general improvements in predictive quality. The evaluation finds remarkably large improvements in the performance metrics for the more sophisticated methodology in comparison to the simple alternatives. This indicates that a more comprehensive assessment of all data preprocessing steps with respect to predictive quality is recommended. Based on the processed data, probabilistic base algorithms are optimized and their predictive performance is assessed to select a suitable candidate underlying the comparison of the multi-label approaches. The results suggest that improvements in predictive performance caused by the optimization are rather small. However, this step identifies regularized Logistic Regression and Gradient Boosting as best probabilistic base algorithms. Where required, the former underlies the multi-label approaches in their comparison. For that, a more sophisticated experimental setup is used. The 5×2 cross-validation yields the expected performance of the approaches and enables statistical tests for significance in performance differences. The benchmark approach is Binary Relevance with Logistic Regression, which does not utilize dependencies between the labels. The majority of multi-label approaches are not able to improve over the benchmark according to the performance metrics. Still, the use of Fully Conditional Dependency, which solely utilizes label dependency, shows improvements in the expected values of the performance metrics. However, the 5×2 paired t-test finds the performance difference to be insignificant at a 5% confidence level. This changes with the additional incorporation of information signals from varying probabilistic base algorithms by the Heterogeneous Bagging Ensemble of Binary Relevance, Heterogeneous Stacking and Fully Conditional Dependency. The additional use of heterogeneity leads to an overall significant improvement in the average Brier Score as strictly proper scoring rule and the average Area Under the Receiver Operating Characteristic curve as ranking metric. Accordingly, this thesis finds that the sole use of label dependence in the considered approaches does not lead to a sufficiently high improvement in discrimination of predictions. Taking further factors like heterogeneity or sophisticated data preprocessing into

account, the predictions are significantly improvable.

However, the performance difference remains insignificant for the Area Under the Precision-Recall Curve. This results from the increased variance which reflects the uncertainty caused by the imbalance in the positive class. Here, the evaluation results fluctuate more due to the higher dependency on the allocation of single positive class instances between each training and test dataset. Further, extreme class imbalance shows a more general adverse effect on precision, making exact identification of the positive class difficult. As illustrated in Figure 6.2, a comprehensive assessment of class imbalances is reasonable in addition to the more detailed analysis of all data preprocessing steps. A possible adjustment to address this imbalance is the use of class weights for the algorithms. This is done by reweighting of the loss function by inverse frequencies of the respective class. With that, the minority class is emphasised and its impact on the empirical risk is increased. A further option to assess class imbalance is over-sampling of the minority class, e.g. with the synthetic minority over-sampling technique by Chawla et al. (2002). A `Python` implementation is made available by Lemaître et al. (2017). Accordingly, over-sampling can be conducted via a decomposition to single disease prediction problems or even within the multi-label framework which is possible by the extensions from Charle et al. (2015). More generally, a decomposition to single targets can be beneficial in terms of label specific data preprocessing. This allows for an explanatory variable selection for each disease target with imputation, encoding and scaling taking place on a target specific dataset. Then the probabilistic predictions are unified in Problem Transformation and Approach Ensembles. Further, improvements are possible with additional Feature Engineering as displayed in Figure 6.2. This involves the application of variable transformations, the use of variable interaction terms and further data manipulation techniques. An intuitive and powerful, yet already available, example is the Body-Mass-Index. More sophisticated techniques for a thorough use of the available information in variables are given in Niculescu-Mizil et al. (2009). Binning and grouping with shallow decision trees are used for an exhaustive information extraction.

As discussed at several stages of the thesis, Calibration of the predictions is a possible next step after improvements in discrimination. For that, Platt Scaling by Platt (1999) can be used. It passes the initial probabilistic prediction through a sigmoid function and minimizes the binary-cross entropy loss with respect to the target. Isotonic Regression by Zadrozny & Elkan (2001) identifies a more general function for calibration that is just required to be monotonically increasing. It is minimized with respect to the Brier Score. Since training the calibration algorithm on the discriminative model training set induces a bias, a holdout calibration set can be used to train and then transform predictions (Niculescu-Mizil & Caruana (2005), p. 2). This is extendable to cross-validation where the discriminative algorithm is fitted per fold to the training data and the calibration algorithm is fitted to the

respective validation data. The transformation on a test set or for out-of-sample predictions results from the averaging over all discrimination and calibration models. Here, calibration is a rescaling that is not intended to change the discriminative performance.

If the focus is on disease classification as a next step, calibration is not obligatory. A perfect classification result is in theory directly attainable by the optimization of discrimination. This can result from any ranking of all predictions for positive class instances above the predictions of negative class instances. However, here as well as for any imperfect discrimination in predictions, the theoretic default threshold for a perfectly calibrated probabilistic classifier of $\tau = 0.5$ in equation 2.1, is generally not optimal. A classification metric of interest can be optimized by threshold moving. This is for instance shown by Fan & Lin (2007) using label dependencies in the multi-label classification framework. Here, the F-Measure, the harmonic mean of Precision and Recall, is optimized. This classification metric is closely related to the *PRC* in Figure 5.8, which visualizes respective combinations for all possible thresholds. Since class imbalance is a major problem, threshold moving simultaneously becomes an option to tackle class imbalance in classification. Neglecting label dependencies, X. Zhang et al. (2020) suggest a simple solution to label imbalance in the context of multi-label classification by a threshold that recovers the respective label distributions of the training set in the test set. More general approaches are possible in the single label framework as for example discussed in Zou et al. (2016). A further optimization of the underlying algorithm and the multi-label approach to the specific classification result is then advisable as well.

Regardless of the followup step, any population level use of models based on the survey data requires a model that has been trained using survey-weighted instances. This ensures that models are representative for the population. Using such models together with the collection of the identified relevant explanatory factors or inferring them from other sources, enables prediction of disease probabilities as well as classification for the population.

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Tobias Flörchinger

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